The Asymmetry of Basis Momentum in Commodity Futures*

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Abstract

Basis momentum, a leading predictor of commodity futures returns, is traded as a symmetric long-short signal. We show that its two signs carry different information. In 43 commodity futures markets over 1960–2024, returns rise after positive basis momentum and, on average, do not fall after negative basis momentum: at the median volatility, one unit of a positive signal raises the expected return by 0.40% per month and one unit of a negative signal by 0.01%, and the difference between the two slopes has a t-statistic of 4.2. The signal carries two kinds of information. Both signs predict the spread between the first two contracts with similar slopes, whereas only positive signals predict the level of the curve, the return its contracts share. A neural network whose specification contains no split or kink recovers the same shape, and two slopes capture what it finds. In the opportunity set, the positive side carries the value, and short positions on negative signals earn no premium. The pattern is not explained by volatility or temporary price pressure and fits the theory of storage, in which a tightening front of the curve signals scarcity. The contrast between level and spread persists after 2010, while the difference between the outright slopes narrows.

Keywords: Basis Momentum, Commodity Futures, Convenience Yield, Hedging Pressure, Spot and Term Premiums, Return Predictability.

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1 Introduction

Basis momentum, the difference between the past returns of the first- and the second-nearby futures contract, is among the strongest predictors of commodity futures returns (Boons and Porras Prado, 2019) and a standard factor in models of the commodity market (Sakkas and Tessaromatis, 2020), next to momentum and the basis (Erb and Harvey, 2006; Szymanowska, de Roon, Nijman, and van den Goorbergh, 2014). Every implementation so far treats the signal as symmetric. A cross-sectional rank replaces its sign with a position relative to other markets, and a linear slope gives a strongly negative value as large a short position as a strongly positive value receives in a long one. Both assume that a nearby contract that has lagged the next one is as informative as one that has led it, and that shorts pay as longs do.

The theory of storage gives a reason to doubt this symmetry. Arbitrage caps a contango at full carry, but nothing caps a backwardation, because scarce inventories cannot be borrowed and sold (Working, 1949; Routledge, Seppi, and Spatt, 2000). The cap concerns the level of the basis, and its implication for basis momentum has not been tested. Earlier evidence points the same way. The low portfolio of Boons and Porras Prado (2019) earns a significantly negative spreading return but no significantly negative nearby return, and in their Internet Appendix, mean nearby returns across 21 markets average about 12% per year after positive and about zero after negative basis momentum. In the cross-section, the coefficient on relative basis is more than three times larger when relative basis is positive (Gu, Kang, Lou, and Tang, 2025). Such comparisons include each market's premium and cannot separate a one-sided from a linear response. This paper estimates the slope on each sign within markets, locates the difference on the curve, and measures what it is worth.

The two signs carry different information. Returns rise after positive basis momentum and, on average, stay flat after negative basis momentum. Across 43 commodity futures over 1960–2024, with each return over its own volatility, the slope is 0.059 above zero (t-statistic

of 6.4) and 0.002 below it (0.2). At the median volatility of a market, one unit of a positive signal raises the expected return by 0.40% per month, and one unit of a negative signal by 0.01%. The difference between the slopes has a t-statistic of 4.2 and survives controls for momentum, the basis, and hedging pressure. The response is strongest for large positive signals, and the change in slope lies between zero and one.

The signal carries two kinds of information, and the futures curve separates them. The spread between the first two contracts responds to both signs, with slopes of 0.020 and 0.034 that do not differ significantly. The level of the curve, the return that its contracts share, responds to positive signals only: above zero, the slopes on the first three contracts are 0.059, 0.061, and 0.063, below zero they are close to zero, and the asymmetry of the level exceeds that of the spread (t-statistic of 4.3). Negative basis momentum thus informs about the shape of the curve and not about its level, whereas positive basis momentum informs about both. Nor is the information the current curve repackaged: with the basis at the time of the signal as a control, the difference between the slopes is unchanged (t-statistic of 4.4).

A flexible estimator finds the same shape without being told where to look. A rank sort cannot show the shape, a linear regression excludes it, and two slopes or a kink assume its form. We therefore estimate the response with a neural network whose only input is the signal and whose specification contains no split, kink, or number of changes in slope. The network recovers the one-sided shape. Its difference between the slopes lies 3.0 standard deviations above its mean under a linear truth (p-value of 0.001), it shows no asymmetry when the truth is a symmetric S-shape, and it detects a one-sided truth in 94% of simulated samples, against 98% for two slopes that know the location. A spline and a kernel regression agree. Machine learning thus serves as a test of the specification, and its verdict is that two slopes capture the shape: once the asymmetric position is in the regression, the network's alpha is -0.14% per month.

The value lies on the positive side. Once positive signals are known, negative signals add

nothing to the variation explained within markets (an increment of 0.000 percentage points of R^2, standard error 0.005), and out of sample they lower it (-0.05 points, t-statistic of -2.0). For an investor who already holds the commodity premium through a long-only portfolio, adding the positive side raises the squared maximum Sharpe ratio by 0.67 (t-statistic of 2.3) and adding the negative side by 0.01, and short positions taken solely because basis momentum is negative earn no premium. A position with separate slopes, estimated on past data, earned an alpha of 0.30% per month gross of costs over 1990–2024 (t-statistic of 3.0), about 1.2% per year at 10% volatility, beyond linear basis momentum and the long-only portfolio, and 0.24% beyond time-series momentum as well. The slopes also differ under the contract rule of Boons and Porras Prado (2019), and Section 6 reports the position with the location of the split and the weights of the benchmarks estimated in real time.

The practical reading is simple: in outright futures, the positive side carries the information; in calendar spreads, both sides do. A short position after a negative signal carries costs and volatility without a premium, whereas the spread responds to both signs, and a linear use of the signal fits there. For hedgers, who are mostly short, a positive signal predicts a higher expected cost of the hedge on the first three contracts, and a negative signal does not predict a lower one. For asset pricing, a factor built from the rank of basis momentum mixes a long leg that carries the premium with a short leg that does not.

The pattern fits scarcity and not two common alternatives. Volatility rises with the size of the signal by similar amounts on both sides of zero, so volatility alone cannot explain a return on one side only, and the gains after positive signals are not reversed over three years, contrary to temporary price pressure. The commodity premium and a symmetric S-shape do not produce the asymmetry either, and the construction of the signal changes its size but not its sign. What remains fits scarcity that the current curve does not yet reflect: positive signals raise the median and above all the upper quantiles of the return, and they are followed by a higher probability of backwardation. Because basis momentum also predicts financial futures, Boons and Porras Prado (2019) and Gu et al. (2025) argue

against a storage channel; the split by sign gives a direct test of that argument.

The contrast between level and spread persists over 2010–2024 (t-statistic of 2.1), while the difference between the outright slopes narrows (t-statistic of 0.8) and the alpha of the position falls to 0.17% per month. The two kinds of information are thus a lasting feature of the curve, and the one-sided premium in outright returns is strongest before 2010.

Our results contribute to three literatures. First, we show that basis momentum carries two kinds of information rather than one. The work on term-structure predictability in commodity markets (Erb and Harvey, 2006; Szymanowska et al., 2014; Koijen, Moskowitz, Pedersen, and Vrugt, 2018; Bakshi, Gao, and Rossi, 2019) establishes that basis momentum predicts returns in linear form, and portfolio legs and conditional means share the direction of our first result (Boons and Porras Prado, 2019). We report the slopes by sign within markets and their contrast along the curve: one-sided in the level, symmetric in the spread. Among related signals, only basis momentum is asymmetric under every common scaling. Relative basis predicts the cross-section more strongly when positive (Gu et al., 2025), whereas our unscaled analogue shows no significant asymmetry within markets (t-statistic of 1.2), and for the traditional basis they find no difference between the signs. In this journal, Luo, Xue, Yu, Li, and Du (2026) construct spot-based versions of the basis and basis momentum for Chinese futures, Zheng, Liu, Wu, and Chen (2026) study curve momentum, as do Paschke, Prokopczuk, and Simen (2020), and Kim (2025) shows that commodity risk factors explain spot and term premiums differently. We keep the numerator of the conventional signal, so that the asymmetry is a property of the established predictor and not of a new construction, and tests of its economic source should be run by sign.

Second, we show which part of a symmetric position earns its premium. Linear and rank positions hold both signs alike, and Maio and Kwon (2024) dispute that basis momentum prices the cross-section of commodity returns; our results concern prediction within markets. In the opportunity set, the positive part of a symmetric position carries its value,

and the negative part earns no corresponding premium. On the six factors of Sakkas and Tessaromatis (2020), the alphas of the two parts differ by 0.83% per month over 1987–2024 (t-statistic of 2.4) and by 0.42% over 1990–2024 (1.4). This differs from the equity evidence, in which the short legs of anomalies are the more profitable after high sentiment, consistent with impediments to short selling (Stambaugh, Yu, and Yuan, 2012): within markets, the information in outright returns lies in positive signals, although a futures position is as easy to sell as to buy. This changes the interpretation of basis momentum from a symmetric long-short signal to a predominantly one-sided predictor.

Third, we use machine learning as a test of form rather than as a forecasting device. Gu, Kelly, and Xiu (2020) and Lim, Zohren, and Roberts (2019) use networks to improve predictions and positions. In commodity futures, Wang and Zhang (2024) forecast returns with machine learning models, and Jia, Simkins, Zhang, and Zhao (2026) extract carry predictors from the term-structure surface with a convolutional network and link them to scarcity in a storage model. Moskowitz, Sabbatucci, Tamoni, and Uhl (2026) estimate the position as a function of past returns in futures markets with a network, a Sharpe objective, and an imposed odd response, which they also relax. Our network targets the conditional mean, leaves asymmetry free, and is held against linear, kink, spline, and kernel forms under one null. It reaches the one-sided shape at little cost in power and confirms that a two-parameter position captures it, so that machine learning here disciplines the specification. Unlike the neglected-nonlinearity tests of White (1989) and Lee, White, and Granger (1993), which return a rejection, it also estimates the shape.

The remainder of the paper proceeds as follows. Section 2 describes the data and the empirical framework. Section 3 establishes the one-sided shape, Section 4 shows that the functional form does not impose it, and Section 5 locates it on the curve. Section 6 measures its economic value, Section 7 confronts it with candidate accounts and reports its weakening since 2010, and Section 8 concludes.

2 Data and Tests of Symmetry

This section describes the data, defines basis momentum and the volatility-scaled return, and sets out the tests of symmetry, the neural network, and the criteria by which we judge statistical and economic significance.

2.1 Commodity Futures Data

The dataset consists of 45 commodity futures in energy, grains and oilseeds, livestock and dairy, industrial metals, precious metals, and softs. For each market, we collect end-of-day settlement prices of the first-, second-, and third-nearby contracts from Bloomberg (Bloomberg L.P., 2024), both roll-adjusted and unadjusted, and the positions of commercial traders from the Commitments of Traders reports of the CFTC, ICE, and the LME (U.S. Commodity Futures Trading Commission, 2024; ICE Futures Europe, 2024; London Metal Exchange, 2024). Prices are in US dollars. For the state tests, we use the VIX and, before 1990, the VXO (Chicago Board Options Exchange, 2026), and the recession dates of the NBER (National Bureau of Economic Research, 2024). Exhibit A1 of the Online Appendix lists the contracts.

Returns are computed from Bloomberg's ratio-adjusted generic first- and second-nearby series. The generics roll on the first trading days of the delivery month in 38 of the 41 signal markets whose roll dates can be recovered, and within the month for WTI, natural gas, and, until about 2014, Brent. A generic's adjusted return uses the new contract from the close before the roll, so that in markets that roll on the first trading day each monthly return comes from one contract. The excess return of a fully collateralized position is

$$r_{i,t+1} = \frac{F_{i,t+1}(T) - F_{i,t}(T)}{F_{i,t}(T)}, \tag{1}$$

where $F_{i,t}(T)$ is the price at the end of month t of the contract on commodity i that expires at T. We denote by $r_{i,t+1} \equiv r^{(1)}_{i,t+1}$ the excess return of the first-nearby , by $r^{(2)}_{i,t+1}$ and $r^{(3)}_{i,t+1}$ those of the second- and third-nearby contracts, and by $F^{(n)}_{i,t}$ the unadjusted price of the n-th nearby contract. Unadjusted prices measure the state of the curve, namely the basis $\ln F^{(1)}_{i,t} - \ln F^{(2)}_{i,t}$ and the relative basis, and isolate the gap at a roll.

Four data screens apply. For the old NYMEX unleaded gasoline contract (1986–2006), whose adjusted series coincides with the unadjusted one, we compute returns from unadjusted prices on the NYMEX roll calendar; lumber, which has no roll-adjusted series, is excluded. Prices that remain unchanged for 20 or more consecutive business days are treated as missing, except for the first day of such a run. Gaps of up to two days are filled with the last price, and never with later prices. Trader positions enter with the last report dated at least seven days before the end of the month in which a position is formed. All signals are therefore known at the end of month t, before the return of month $t+1$ is earned.

The number of markets grows from 6 in 1960 to 30 in 1990 and 41 in 2010 (Exhibit A3 and Exhibit A2 of the Online Appendix). Because the signal requires almost two years of returns, the first predictive regression month is July 1961. We refer to the regression sample as 1960–2024. It ends in June 2024, the last full month of the adjusted price data, which rest on one consistent Bloomberg vintage.

2.2 Basis Momentum and Its Two Signs

Basis momentum is the difference between the 12-month cumulative returns of the first- and the second-nearby contract, the numerator of Boons and Porras Prado (2019):

$$BM_{i,t} = \prod_{k=t-11}^{t} \left(1 + r_{i,k}\right) - \prod_{k=t-11}^{t} \left(1 + r^{(2)}_{i,k}\right). \tag{2}$$

It is available for 43 of the 45 markets, because propane and pork bellies lack an adjusted second-nearby series. It differs from price momentum, which measures the level of past returns, and from the basis, which measures the current slope of the curve. We apply it to Bloomberg generics, which hold the nearby contract until its delivery month, whereas Boons and Porras Prado (2019) hold contracts that expire after $t+2$. Exhibit A13 of the Online Appendix compares the two. As controls and comparison signals, we use 12-1 month momentum, the basis, the relative basis $\ln F_{i,t}^{(1)} - 2\ln F_{i,t}^{(2)} + \ln F_{i,t}^{(3)}$, and the hedging pressure of commercial traders, (short − long)/(long + short); Section G of the Online Appendix gives their scaling.

Because a cross-sectional rank removes the sign of the signal, our question requires its time-series version. We standardize each market's signal by its own volatility and express each market's return per unit of its ex-ante volatility:

$$z_{i,t} = \frac{BM_{i,t}}{\hat{\sigma}_{12}(BM)_{i,t}}, \qquad \tilde{r}_{i,t+1} = \frac{r_{i,t+1}}{\hat{\sigma}_{12}(r)_{i,t}}, \tag{3}$$

where $\hat{\sigma}_{12}$ is the standard deviation over the trailing 12 months, so that both quantities are known at the end of month t. We call $z_{i,t} > 0$ positive and $z_{i,t} < 0$ negative basis momentum, with parts $z^+ = \max(z, 0)$ and $z^- = \min(z, 0)$. The unit of $z_{i,t}$ is a market's own trailing standard deviation of BM. The pooled standard deviation of the signal is 2.31, and a signal above two occurs in 16% and a signal below minus two in 19% of the observations.

The theory of storage gives the two signs different economic content. Basis momentum is positive when the nearby contract has outperformed the next, so that the front of the curve has tightened, as it does when inventories are drawn down; it is negative when the front has weakened relative to the next contract, as it does when inventories accumulate. Two predictions follow (Working, 1949; Brennan, 1958; Routledge et al., 2000; Gorton, Hayashi, and Rouwenhorst, 2013). When inventories are ample, arbitrage bounds the curve at full carry, so a weakening front should inform the spread between contracts rather than the level

of the curve. When inventories are scarce, the convenience yield is convex in them, so a tightening front should raise the expected return of every contract that matures before new supply arrives, and above all the upper tail. These predictions concern the level of the basis, and no formal model carries them to basis momentum; we use them to read the results, not to identify their cause.

2.3 Tests of Equal Information in Both Signs

Our object of interest is the expected volatility-scaled return given the signal, net of each market's commodity premium, $g(z) = E[\tilde{r}_{i,t+1} - \bar{\tilde{r}}_i \,|\, z_{i,t} = z]$, where $\bar{\tilde{r}}_i$ is the mean scaled return of market i. Under the null hypothesis, g is linear, and the two signs carry the same information. We start from the linear benchmark

$$\tilde{r}_{i,t+1} = a_i + \beta\, z_{i,t} + \varepsilon_{i,t+1}, \tag{4}$$

where the market fixed effects a_i absorb each market's average return, and relax it in the asymmetric regression, which allows a separate slope on each side of zero:

$$\tilde{r}_{i,t+1} = a_i + \beta^{+} z^{+}_{i,t} + \beta^{-} z^{-}_{i,t} + \varepsilon_{i,t+1}. \tag{5}$$

A Wald test of $\beta^{+} = \beta^{-}$ is the central test of the paper. Zero is a natural reference point, but imposing it could create the result. We therefore estimate where the slope changes with a regression kink at an unknown threshold (Hansen, 2017),

$$\tilde{r}_{i,t+1} = a_i + \phi\, z_{i,t} + \delta\, (z_{i,t} - \gamma)_{+} + \varepsilon_{i,t+1}, \tag{6}$$

which is continuous, with slope ϕ below the kink γ and $\phi + \delta$ above it. We estimate γ by minimizing the sum of squared residuals over a grid between the 5th and 95th percentiles of

the signal.

The same regression locates the information along the curve. The nearby return is the return of the second-nearby contract, which we call the level, plus the spreading return, long the first- and short the second-nearby contract:

$$r^{(1)}_{i,t+1} = r^{(2)}_{i,t+1} + s_{i,t+1}, \qquad s_{i,t+1} = r^{(1)}_{i,t+1} - r^{(2)}_{i,t+1}. \tag{7}$$

We estimate Equation (5) for each component $c \in \{L, S\}$, the level $r^{(2)}$ and the spread s, each over its own trailing volatility, and write the asymmetry of a component as $\Delta_c = \beta_c^+ - \beta_c^-$. Two hypotheses follow: $\Delta_L = 0$, under which the level responds to both signs alike, and $\Delta_L - \Delta_S = 0$, under which the level and the spread share the asymmetry. We test the second in a stacked regression of both components with fixed effects by market and component (Equation (A4) of the Online Appendix). Over the volatility of the nearby return instead, the components add up, and so do the slopes, $\beta^{\pm} = \beta_L^{\pm} + \beta_S^{\pm}$.

A kink still assumes one change in slope. To remove this assumption, we estimate g directly with a neural network whose specification contains no split, kink, or number of changes in slope. The signal enters on its signed scale, with its origin at zero, so the network estimates the shape without an imposed threshold and does not discover zero by itself. The design follows from the object of interest. The signal is the only input, because the question concerns the form in the signal and not its interaction with other variables. The target is net of the market's mean in the training window, because g is net of the premium, and the loss is the mean squared error, because g is a conditional expectation. The intercept is free, because returns net of the market's mean are negative at signals near zero. The architecture is small, because one input and a signal-to-noise ratio of well under 1% call for few parameters: one or two hidden layers with 4 or 8 tanh nodes, trained on a pooled panel of all markets. With

one hidden layer of K nodes, the fitted response and the loss are

$$\hat{g}(z) = b_2 + \sum_{k=1}^{K} w_{2,k} \tanh\big(w_{1,k} z + b_{1,k}\big), \qquad \min_{\theta} \frac{1}{N} \sum_{i,t} \big(\tilde{r}_{i,t+1} - \bar{\tilde{r}}_i - \hat{g}(z_{i,t};\theta)\big)^2 + \omega \|\theta_w\|^2, \quad (8)$$

where $\bar{\tilde{r}}_i$ is the market's mean in the training window and ω penalizes the weights θ_w; a second layer applies the same map to the K outputs. We select among 32 configurations on the last four years of each training window and average ten seeds of the selected configuration. We fit the full sample for the shape and use expanding windows with data through December 1989, 1997, 2005, 2013, and 2021 for forecasts and positions, so that every input is known at the estimation date.

The test statistic is the difference between the slopes of the fitted curve, in units of the return: we evaluate $\hat{g}$ on a grid of signals z_j from -4 to 4, regress $\hat{g}(z_j) = c + b^+ z_j^+ + b^- z_j^- + u_j$, and use $\Delta_{NN} = \hat{b}^+ - \hat{b}^-$. Its null distribution comes from simulated returns that keep the real signals: the fitted value of a linear regression plus the residual times one normal draw per month (Equation (A3) of the Online Appendix), in 999 draws, each estimated with ten seeds. Power comes from the same procedure under a one-sided truth, in 199 draws, and specificity from a symmetric S-shape, an odd response that flattens for large signals of either sign. A restricted cubic spline, a kernel regression, an estimated kink, and two slopes run through the same design, so that all estimators face one null. Section C of the Online Appendix gives the full specification, the reason for each choice, and a study of alternative designs.

2.4 Statistical and Economic Criteria

Standard errors are clustered by month to account for the cross-sectional correlation of commodity returns (Petersen, 2009). Because many signals have been tested on commodity returns, we hold the central test to the hurdle of a t-statistic of three (Harvey, Liu, and Zhu, 2016). For the kink, we test $\delta = 0$ with the supremum of the F-statistic over the grid

and obtain the p-value from a fixed-regressor wild bootstrap by month (Hansen, 1996), with an interval for γ from a stationary month bootstrap (Politis and Romano, 1994). Time-series alphas use Newey and West (1987) standard errors with six lags, and regressions over overlapping horizons use Driscoll and Kraay (1998) standard errors.

We keep statistical and economic significance apart. Statistical criteria state whether an effect is distinguishable from zero or a form from another: t-statistics, the out-of-sample R^2 against the prevailing mean of each market, the tests of Clark and West (2007) and Diebold and Mariano (1995), superior predictive ability (Hansen, 2005), and the model confidence set (Hansen, Lunde, and Nason, 2011). Economic criteria state what an effect is worth: alpha, the Sharpe ratio, and the certainty equivalent at a relative risk aversion of 5; Section G of the Online Appendix defines each criterion. A strategy holds a position $w_{i,t} = f(z_{i,t})$ in each market, rebalances monthly, and earns

$$r^p_{t+1} = \frac{1}{N_t} \sum_{i=1}^{N_t} f(z_{i,t})\, \sigma^{target}\, \tilde{r}_{i,t+1}, \tag{9}$$

where $\sigma^{target} = 40\%/\sqrt{12}$ per month, 40% per year (Moskowitz, Ooi, and Pedersen, 2012), because $\tilde{r}_{i,t+1}$ is scaled by a monthly standard deviation, and N_t is the number of markets with a signal. We compare five positions: linear basis momentum, $f(z) = z$; the asymmetric position, $f(z) \propto \hat{\beta}^+ z^+ + \hat{\beta}^- z^-$, with the slopes of Equation (5) estimated on past data; the positive-only rule, $f(z) = \max(z, 0)$, which requires no estimation; the long-only portfolio, $f(z) = 1$, which earns the commodity premium; and time-series momentum, the sign of a market's 12-month return skipping the most recent month (Moskowitz et al., 2012). Estimated positions are rescaled so that their root mean square in the estimation sample equals that of z. Alphas are intercepts of time-series regressions on benchmark returns,

$$r^p_{t+1} = \alpha + b_1 r^{lin}_{t+1} + b_2 r^{LO}_{t+1} + \mathbf{c}^\top \mathbf{x}_{t+1} + e_{t+1}, \tag{10}$$

where r^{lin} is linear basis momentum, r^{LO} the long-only portfolio, and $\mathbf{x}$ further traded benchmarks, such as time-series momentum or the six factors. To measure convexity in the commodity market, we add $\theta\,(r^{LO}_{t+1})^2$ (Treynor and Mazuy, 1966) or $\theta \max(r^{LO}_{t+1}, 0)$ (Henriksson and Merton, 1981). Because these terms are not traded returns, the intercept is then net of the convex payoff and is not an alpha (Merton, 1981).

To measure what each sign contributes, we report each part of the signal alone and both together, and the increment of adding one part to the other. For forecasts, the increment is the loss of R^2 when a part is removed, the variable importance of Gu et al. (2020). For positions, it is the gain in the squared maximum Sharpe ratio when a part is added to the investment opportunity set (Barillas and Shanken, 2017); with the long-only portfolio in every set, the gain measures what a side adds for an investor who already holds the commodity premium. Standard errors come from a delete-one-year jackknife, which keeps the dependence within a year. Shapley values (Shapley, 1953; Lindeman, Merenda, and Gold, 1980), the average of a part's value alone and its increment, give the same ranking and are reported in the Online Appendix.

3 Only Positive Basis Momentum Predicts Outright Returns

This section establishes the shape. We move from a factor model, which mixes the two signs, to sorted returns and slopes within markets, which separate them, then ask where the slope changes and whether established commodity characteristics account for the difference.

3.1 Factor Legs Mix the Two Signs

We begin with the commodity factor model of Sakkas and Tessaromatis (2020): the average of all markets and top-minus-bottom 30% sorts on momentum, the basis, basis momentum, hedging pressure, and value. Panel A of Exhibit 1 shows why basis momentum deserves attention: its factor earns 9.57% per year over 1987–2024 (t-statistic of 3.8), of which 6.24% comes from the long leg and 3.33% from the short leg. A rank sort cannot say more about the two signs, because a market in the short leg may have a positive signal, so that its short-leg return is no test of negative signals.

Exhibit 1: **Basis Momentum in a Commodity Factor Model**

Panel A: BM factor, % per year, 1987–2024	Factor	Long leg	Short leg			
Mean	9.57***	6.24***	3.33**			
	(3.81)	(4.21)	(2.37)			
	Linear	Positive	Negative	Binary	Long-or-flat	Long-only
Panel B: Time-series positions, 1987–2024						
Sharpe ratio	0.60***	0.98***	−0.18	0.58***	0.71***	0.51***
α, six factors	0.44	0.64***	−0.19	0.17	0.12	0.08
	(1.35)	(3.55)	(−0.67)	(1.53)	(1.55)	(0.68)
α, without BM	0.56	0.68***	−0.12	0.22	0.15	0.08
	(1.42)	(3.49)	(−0.37)	(1.54)	(1.64)	(0.63)
α, Positive − Negative	0.83**					
	(2.35)					
Panel C: Time-series positions, 1990–2024						
Sharpe ratio	0.63***	0.85***	−0.10	0.58***	0.64***	0.43**
α, six factors	0.50	0.46***	0.04	0.17	0.07	−0.04
	(1.47)	(2.68)	(0.15)	(1.45)	(0.82)	(−0.39)
α, without BM	0.65*	0.53***	0.12	0.24*	0.10	−0.04
	(1.68)	(2.68)	(0.44)	(1.65)	(1.07)	(−0.39)
α, Positive − Negative	0.42					
	(1.45)					

This table reports the factors of Sakkas and Tessaromatis (2020) and positions that estimate nothing. Panel A: BM factor, top minus bottom 30% of the markets, with long leg minus the average market and average market minus short leg. Panels B and C: Linear $f(z) = z$, Positive $\max(z, 0)$, Negative $\min(z, 0)$, Binary $\text{sign}(z)$, Long-or-flat $\mathbb{1}\{z > 0\}$, with α in % per month on the six factors and on the five without BM; α, Positive − Negative is on the six factors, gross of costs, at the volatility target of Equation (9). Newey–West t-statistics in parentheses. Stars on Sharpe ratios are one-sided, from a bootstrap. ***, **, and * denote significance at the 1%, 5%, and 10% levels.

Time-series positions, which keep the sign, reveal the split. On the six factors, linear basis momentum has an insignificant alpha of 0.44% per month (t-statistic of 1.4). Its positive

part has an alpha of 0.64% (3.5), and its negative part of −0.19% (−0.7), with Sharpe ratios of 0.98 and −0.18, against 0.60 for the linear position; the alphas differ by 0.83% (t-statistic of 2.4). Over 1990–2024, the alphas are 0.46% (2.7) and 0.04% (0.15), and the difference of 0.42% is not significant (1.4). Binary and long-or-flat rules, which do not size positions, have no significant alpha. Because the positive part is net long and each market's mean is still in the return, the next tests estimate the shape within markets.

3.2 Flat Returns across Negative Signals

Exhibit 2 sorts all market-months into quantile bins of the signal and plots the mean volatility-scaled return of the following month, net of each market's mean return. The relation is not linear. Across the bins of negative signals, expected returns show no trend, and none of these bins lies significantly above the market mean. The bins stay level up to about 1.2 and rise above that, and the three highest bins lie significantly above the market mean.

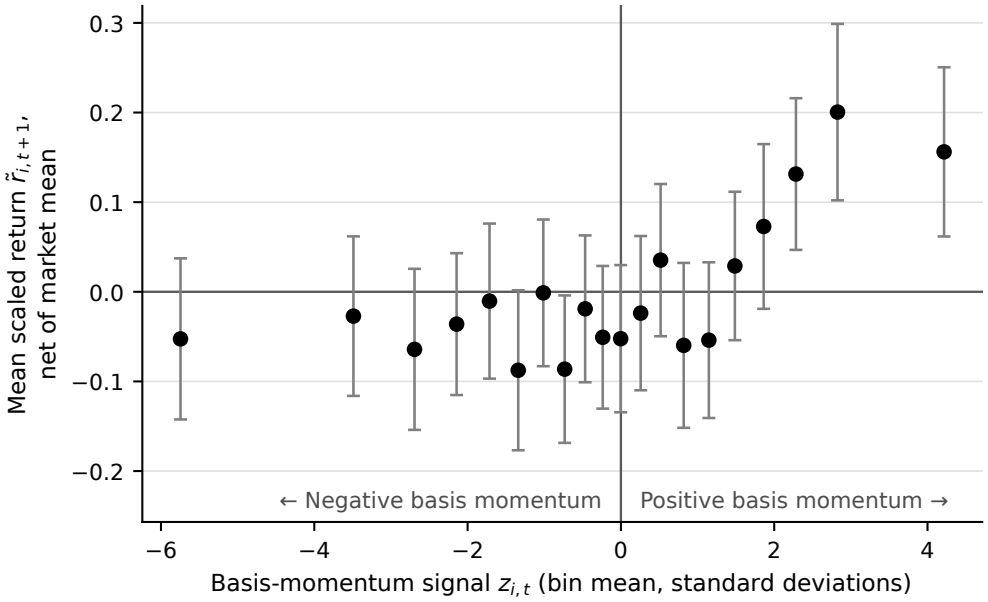


Exhibit 2: **Next-Month Returns Are Flat across Negative Signals and Rise with Large Positive Signals.** Mean of $\tilde{r}_{i,t+1}$ net of the market's mean in 20 quantile bins of $z_{i,t}$, 43 markets, July 1961–June 2024, with 95% confidence intervals clustered by month.

The magnitudes are large. Futures earn 0.07% in the month after a signal of −2 or below, 0.12% and 0.15% in the next two ranges up to zero, 0.28% between zero and one, 0.72% between one and two, and 1.71% above two; net of each market's mean, this last range earns 1.09% (t-statistic of 4.4; Panel D of Exhibit 4).

3.3 A Positive Slope and a Flat Negative Branch

Exhibit 3 turns the picture into a test. The slope on positive basis momentum is 0.059 (t-statistic of 6.4), and the slope on negative basis momentum is 0.002 (t-statistic of 0.2). A rise of one unit in a positive signal is followed by a return higher by 5.9% of the monthly volatility. On the negative side, the 95% interval runs from −0.013 to 0.017 and excludes the linear slope of 0.025. The flat negative branch is an average over the sample. Before 2000, the negative slope between −3 and zero is 0.044 (t-statistic of 2.6), so the asymmetry of that period rests on the most negative signals; in 2000–2009, it is −0.064 (−2.3), a change with a t-statistic of −3.3, while the positive slope does not change; and with a 24-month volatility, the negative branch falls (−0.026, significant at the 5% level; Exhibit A9 and Exhibit A12 of the Online Appendix).

The difference between the slopes clears the hurdle of three over the full sample (t-statistic of 4.2). It remains significant with standard errors clustered by month and market (4.7) and with Driscoll and Kraay (1998) standard errors (3.8), and with month fixed effects (Wald p-value of 0.003, and 0.09 after 2000). It lies between 3.9 and 4.4 without any one market and between 3.7 and 4.4 without any one calendar year, and it clears the hurdle in 37 of 57 constructions and samples of the signal (Exhibit A4 and Exhibit A5 of the Online Appendix). A mean-group estimate, which weights every market equally whatever the length of its history, has a t-statistic of 2.0. It holds with 6- or 24-month basis momentum, with longer volatility windows, with an unscaled signal, without energy, precious metals, or crises, and in the 24 markets of the S&P GSCI, and the slope on positive basis momentum is positive

Exhibit 3: **Predictive Regressions of Volatility-Scaled Returns on Basis Momentum**

	1960–2024	1960–1999	2000–2024	2010–2024
Linear: $\tilde{r}_{i,t+1} = a_i + \beta\, z_{i,t} + \varepsilon_{i,t+1}$				
β	0.025***	0.040***	0.011	0.018**
	(5.07)	(5.39)	(1.62)	(2.08)
Asymmetric: $\tilde{r}_{i,t+1} = a_i + \beta^+ z^+_{i,t} + \beta^- z^-_{i,t} + \varepsilon_{i,t+1}$				
β^+	0.059***	0.087***	0.040***	0.028**
	(6.39)	(5.63)	(3.40)	(1.99)
β^-	0.002	0.011	−0.011	0.009
	(0.23)	(0.99)	(−0.95)	(0.66)
$\beta^+ - \beta^-$	0.057***	0.076***	0.051***	0.018
	(4.16)	(3.53)	(2.74)	(0.82)
Wald $\beta^+ = \beta^-$, p-value	0.000	0.000	0.006	0.411
$t(\beta^+ - \beta^-)$, clustered by month and market	4.70	3.72	3.10	0.96
$t(\beta^+ - \beta^-)$, Driscoll–Kraay	3.79	3.29	2.73	0.92
Asymmetric with month fixed effects				
β^+	0.046***	0.072***	0.029***	0.021
β^-	0.009	0.014	0.001	0.007
Wald $\beta^+ = \beta^-$, p-value	0.003	0.003	0.091	0.522
Curvature: $\tilde{r}_{i,t+1} = a_i + \beta\, z_{i,t} + \beta_3\, z^3_{i,t} + \varepsilon_{i,t+1}$				
$\beta_3 \times 1{,}000$	−0.10	−0.72***	0.13	0.02
	(−0.84)	(−3.31)	(1.00)	(0.16)
Within R^2 (%), linear / asymmetric	0.21 / 0.33			
Markets	43	36	43	43
Observations	19,327	7,860	11,467	7,160

This table reports predictive regressions of $\tilde{r}_{i,t+1}$ on $z_{i,t}$ (Equation (3)) with market fixed effects, monthly; the row Markets gives the markets. β_3 is multiplied by 1,000. t-statistics in parentheses are clustered by month. ***, **, and * denote significance at the 1%, 5%, and 10% levels.

in all six sectors (Section B of the Online Appendix).

The construction of the signal changes the size of this result but does not produce it. We rebuild the signal and the scaled return from panels in which both return series are flipped about the market's mean by one sign per month, which keeps the premium, the mean spread, and the clustering of volatility. The t-statistic of the difference averages 0.05, with a maximum of 2.94 in 300 draws. When returns are truly linear in other transformations of the signal, at most 0.5% of the draws reach the observed value (Exhibit A7 of the Online Appendix). The numerator matters for size. With a log numerator, the difference keeps its sign and has a t-statistic of 2.3, below our hurdle. Because basis momentum equals the past gross return times twice the hyperbolic sine of half the log numerator, the two forms weight large values differently; with the past return as a control, the t-statistic is 3.3 (Exhibit A13

of the Online Appendix).

3.4 A Change in Slope between Zero and One

The regression kink confirms that the slope changes without imposing where (Exhibit 4; Exhibit A10 of the Online Appendix plots the F-statistic over the grid). None of 999 bootstrap draws exceeds the observed sup-F statistic. The estimated location is -0.10, with a slope of 0.001 below the kink and 0.058 above it (t-statistic of 5.5), and half of all observations lie above it. The estimate thus lies near zero, within a wide interval. With a kink fixed at -1, 0, and 1, the t-statistic of the change in slope is 3.9, 4.2, and 3.9, with almost the same fit, and the 90% interval of the location runs from -2.90 to 0.95. Rejecting linearity is thus easier than dating the change, and zero is the natural benchmark within the interval. The rejection draws on large signals: without signals above two, 16% of the observations, the difference is 0.046 (t-statistic of 1.9). The data place the change between about zero and one and do not separate a kink from a rise concentrated in large positive signals.

3.5 Not a Proxy for Known Characteristics or Sectors

The asymmetry is not a proxy for the established commodity characteristics. Panel A of Exhibit 5 adds the positive and negative parts of momentum, the basis, and hedging pressure, in their common sample from 1986. The slope on positive basis momentum falls at most from 0.054 to 0.044 and stays at 0.047 (t-statistic of 4.1) with all three controls, the slope on negative basis momentum stays at zero, and the difference remains significant in every column (t-statistics between 2.9 and 3.6). Panel B compares the two signs within terciles of each control. In all nine terciles, returns after positive signals exceed those after negative signals, and the positive-side slope exceeds the negative-side slope. The result is also specific to basis momentum: it is significantly asymmetric under all six common scalings of the signal,

Exhibit 4: **Regression Kink Estimates of the Basis-Momentum Relation**

	$\hat{\gamma}$	90% interval	Slope below	Slope above	p (no kink)	BM at kink (%)	Share above (%)
Panel A: Full sample and subperiods							
1960–2024	−0.10	[−2.90, 0.95]	0.001	0.058***	<0.001	−0.2	50
1960–1999	−2.90	[−4.10, 0.90]	−0.039	0.063***	0.001	−7.2	88
2000–2024	0.20	[−1.58, 1.50]	−0.010	0.043***	0.013	0.4	46
Panel B: Expanding estimation windows							
1960–1989	−3.00	[−3.80, 3.16]	−0.029	0.060***	0.063	−7.8	86
1960–1997	−2.90	[−4.21, 0.40]	−0.038	0.064***	0.001	−7.3	88
1960–2005	−0.30	[−2.81, 0.90]	0.001	0.075***	0.002	−0.7	48
1960–2013	−0.20	[−2.71, 0.20]	−0.004	0.068***	<0.001	−0.5	49
1960–2021	−0.10	[−2.81, 0.90]	−0.003	0.059***	<0.001	−0.2	49
Panel C: Storability							
Storable	−0.10	[−2.75, 0.95]	0.003	0.061***	0.001	−0.2	50
Precious metals	−0.30	[−2.95, 2.80]	−0.029	0.061**	0.125	−0.1	47
Livestock and dairy	2.60	[−3.70, 3.15]	0.009	0.096	0.411	10.0	9

Panel D: Excess return in the month after the signal, % per month, simple means

Signal	Mean	Net of market mean	t	Share (%)
$z_{i,t} \leq -2$	0.07	−0.25	−1.27	19
$-2 < z_{i,t} \leq -1$	0.12	−0.28	−1.32	14
$-1 < z_{i,t} \leq 0$	0.15	−0.34	−1.60	20
$0 < z_{i,t} \leq 1$	0.28	−0.25	−1.19	18
$1 < z_{i,t} \leq 2$	0.72	0.19	0.86	14
$z_{i,t} > 2$	1.71	1.09	4.40	16

This table reports Equation (6) with market fixed effects, monthly. Panels A and C cover the stated sample through June 2024; Panel B uses expanding windows through December of the final year. $\hat{\gamma}$ is in units of $z_{i,t}$, with a 90% stationary-bootstrap interval; p is from the sup-F test with a wild bootstrap by month. BM at kink is the median of $\hat{\gamma}\hat{\sigma}_{12}(BM)_{i,t}$. Panel D reports simple mean excess returns in the following month, 43 markets, with the t-statistic of the net mean clustered by month. ***, **, and * denote significance at the 1%, 5%, and 10% levels.

whereas momentum and the basis are so only under their own rolling volatility, where the construction itself tends to produce it (Exhibit A11 of the Online Appendix). Under the baseline scaling, the asymmetry of basis momentum does not differ significantly from that of momentum or the basis (t-statistics of 0.2 and 0.6). It is also not a sector effect: split each month into its sector mean and the deviation from it, both parts of the signal are asymmetric (t-statistics of 2.2 and 3.1), and the two asymmetries do not differ significantly (Panel B of Exhibit A6 of the Online Appendix). Our unscaled analogue of relative basis shows slopes of 0.029 and −0.009 (t-statistic of 1.2), a contrast with the cross-sectional result of Gu et al. (2025) that we do not read as a contradiction, because their measure is time-scaled and their design differs. Every test that separates the two signs within markets thus finds a one-sided response, on the side on which the front of the curve tightens, as the theory of storage leads one to expect. Each of them, however, imposes a split at zero or allows a single change in

slope, and Section 4 removes that assumption.

Exhibit 5: **Asymmetric Regressions with Momentum, Basis, and Hedging-Pressure Controls**

Panel A: Asymmetric regressions with controls

	BM only	+ Momentum	+ Basis	+ HP	+ All
β^+	0.054***	0.044***	0.054***	0.052***	0.047***
	(4.83)	(3.87)	(4.81)	(4.67)	(4.15)
β^-	−0.005	−0.005	−0.006	−0.003	−0.004
	(−0.47)	(−0.51)	(−0.55)	(−0.34)	(−0.35)
$t(\beta^+ - \beta^-)$	3.53	2.93	3.59	3.34	3.04
Observations	12,240	12,240	12,240	12,240	12,240

Panel B: Double sorts on the sign of z and terciles of the control

		Mean $\tilde{r}_{i,t+1}$, net of market mean			Regression within tercile		
Control	Tercile	$z > 0$	$z < 0$	Difference	β^+	β^-	$t(\beta^+ - \beta^-)$
Momentum	Low	−0.021	−0.068	0.048 (1.28)	0.020	0.014	0.21
	Mid	0.038	−0.065	0.103*** (3.66)	0.080***	−0.009	3.66
	High	0.094	0.030	0.064* (1.83)	0.054***	−0.009	2.58
Basis	Low	0.016	−0.067	0.083** (2.48)	0.047**	0.007	1.68
	Mid	0.019	−0.029	0.048 (1.52)	0.050***	0.007	1.69
	High	0.087	−0.030	0.117*** (3.44)	0.071***	−0.036*	3.90
Hedging pressure	Low	0.024	−0.026	0.050 (1.20)	0.035*	−0.003	1.33
	Mid	0.006	−0.028	0.034 (0.85)	0.045**	−0.006	1.70
	High	0.052	−0.035	0.087** (2.33)	0.065***	−0.002	2.48

This table reports Equation (5) with controls and double sorts, monthly, with market fixed effects. Panel A: February 1986–June 2024, 38 markets, adding the positive and negative parts of each control, or of all three (+ All). Panel B: sorts into terciles of the control within each month, mean $\tilde{r}_{i,t+1}$ net of the market mean by the sign of the signal. Controls are defined in Section G of the Online Appendix. t-statistics are clustered by month. ***, **, and * denote significance at the 1%, 5%, and 10% levels.

4 Flexible Estimators Recover the Same Shape

Formal tests reject the linear form: the RESET test has a p-value of 0.018 and the neural-network test of Lee et al. (1993) of 0.001, both driven by the period before 2000 (Exhibit A15 of the Online Appendix). This section asks whether the one-sided shape survives estimators that impose neither a split nor a single change in slope, and whether such a flexible form adds anything to two slopes. Section 4.1 reports the neural network, and Section 4.2 compares it with the other forms on the shape, on forecasts, and as a position.

4.1 A Neural Network without an Imposed Split

The asymmetric regression imposes a change at zero, and the kink allows one change. The specification of the network contains no split, kink, or number of changes in slope, so it tests the shape from the opposite direction. It recovers the shape (Exhibit 6). In units of the return, its slope is 0.057 for positive and 0.007 for negative signals, a difference of 0.050, against 0.066 for two slopes under the same design.

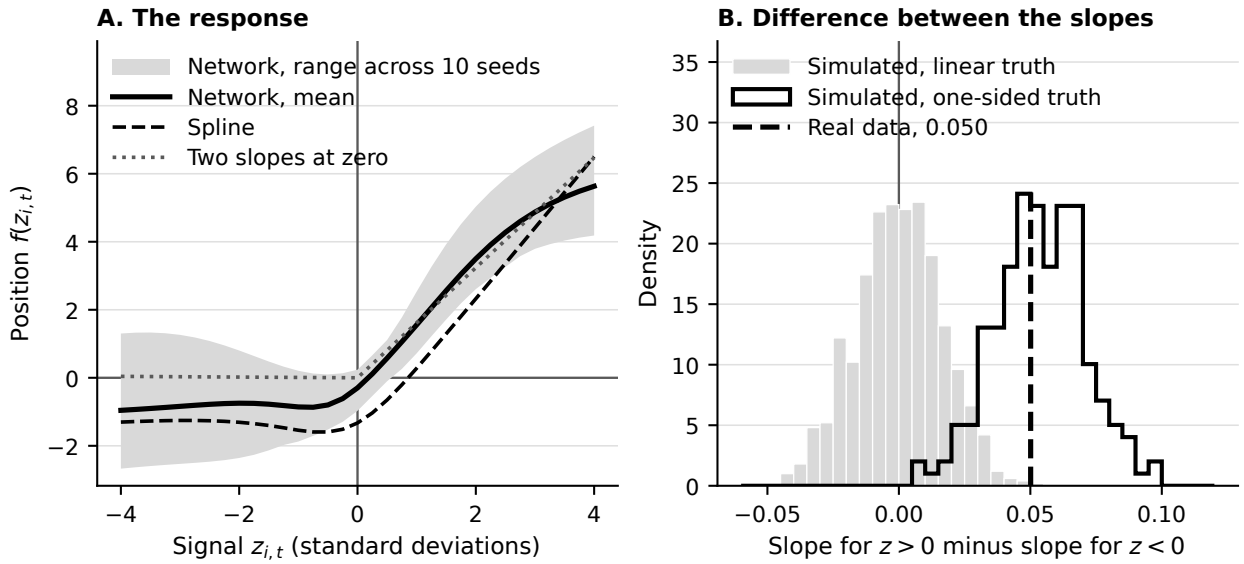


Exhibit 6: **Response Estimated by the Neural Network.** Panel A: position $f(z_{i,t})$ for signals between −4 and 4, July 1961–June 2024, for the network, a spline, and two slopes at zero. Panel B: difference between the slopes of the network, in units of the return, in 999 simulated draws under a linear and 199 under a one-sided truth, and on the real data.

The network does not invent this shape. When returns are simulated as truly linear in the signal, the difference averages zero with a standard deviation of 0.016, so that the observed value lies 3.0 standard deviations above it (p-value of 0.001, at the floor of 999 draws; 0.03 for the scaled difference). When they are truly one-sided, it averages 0.054, and the network detects them in 94% of the draws; on the scaled difference, which also reflects the size of the response, detection in the smaller design study is 66% (Exhibit A21 of the Online Appendix). Under a truly symmetric S-shape, it shows no asymmetry (a scaled difference of 0.04, against 1.48 on the real data). The result also does not hinge on one design:

across eleven alternatives in a study fixed in advance, the scaled difference stays between 1.15 and 1.92, against 1.48 in the main specification, except with dropout, which flattens any response. With past data, the scaled difference is 2.61 and 3.36 with data through 2005 and 2021 and 0.58 through 1989, when a kink is estimated near -3 (Exhibit A23 of the Online Appendix). It is 0.11 and 0.50 through 1997 and 2013, the two windows in which validation selects dropout (Exhibit A16 of the Online Appendix). With a validation sample of eight years, which selects no dropout in these windows, the median across windows is 1.86 (Exhibit A18); we report this as a reading of the windows and keep the main specification.

4.2 Two Slopes Summarize the Flexible Forms

Exhibit 7 sets the network beside the other forms. The left panel concerns the shape. The linear form cannot show an asymmetry, by construction. Each estimator run through the simulations finds one on the real data, and for each, none of 999 draws under a linear truth reaches the observed value. They differ in what they require and in what they report when the shape is absent or lies elsewhere. Two slopes know the location and show a biased mean difference under a symmetric S-shape (0.29), whereas the network (0.04), the kernel regression (0.02), and the spline (-0.08) do not. The specification of the network has no split, knots, or bandwidth. It tests the shape and does not date the change, because its location estimate leans toward the origin of its input: in 50 draws, it places a true kink at zero within 0.5 in 84% of the cases, and it also says near zero in 58% when the true kink is at 1.2. The spline does so in 54% and 12%, and it locates the kink at 1.2 in 88%, against 32% for the network. The network is also further from the true curve on average (Panel C of Exhibit A22 of the Online Appendix). Where the forms predict, they find the same signal: the departures of the forecasts from the prevailing mean correlate with those of two slopes at 0.86 for the network and at 0.97 and 0.96 for the spline and the estimated kink, and in simulations the network recovers 70% to 86% of the variation of the true curve, against 83%

to 95% for two slopes (Exhibit A15 of the Online Appendix). As a position, it is the weakest of the flexible forms (alpha of 0.26% per month, t-statistic of 1.5).

Exhibit 7: **The Shape across Estimators**

	Shape, full sample					Forecasts and positions, 1990–2024				
	Slope difference	p, linear truth	Detection	Difference, S-shape	Kink at 0 dated	R^2_{OOS} (%)	CW	DM	CE (%)	ΔCE (p)
Linear						0.20			2.9	
Kink, location estimated					0.44	0.26	3.0	1.0	5.2	2.3 (0.49)
Two slopes at zero	0.066	0.001	0.98	0.29	1.00	0.37	4.7	2.6	6.3	3.4 (0.05)
Kernel regression	0.046	0.001	0.94	0.02	0.34	0.15	2.9	0.1	5.0	2.1 (0.04)
Restricted cubic spline	0.067	0.001	0.97	−0.08	0.54	0.31	3.6	1.6	5.7	2.8 (0.02)
Neural network	0.050	0.001	0.94	0.04	0.84	0.32	2.7	1.3	4.9	2.0 (0.15)

This table compares estimators of the response of $\tilde{r}_{i,t+1}$ to $z_{i,t}$, monthly, 43 markets. Slope difference: slope for $z > 0$ minus that for $z < 0$ of the fitted curve, in units of the return. p, linear truth: one plus the number of 999 draws under a linear truth that reach it, over 1,000. Detection: share of 199 draws under a one-sided truth above the 95th percentile under the linear truth. Difference, S-shape: the mean of the same difference, scaled to the root mean square of the signal, when the truth is a symmetric S-shape, 50 draws. Kink at 0 dated: share of 50 draws with a true kink at zero that locate it within 0.5. R^2_{OOS} is against the prevailing market mean; CW and DM are Clark–West and Diebold–Mariano t-statistics against linear; CE is the certainty equivalent at a risk aversion of 5 and 10% volatility known in real time, in % per year, with the gain over linear and its bootstrap p-value.

The right panel concerns forecasts. Out of sample, the R^2 is 0.20% for the linear regression, 0.26% for the estimated kink, 0.15% for the kernel regression, 0.31% for the spline, and 0.32% for the network, the highest among the forms without a known location, against 0.37% for two slopes at zero. These gains come from the 1990s: since 2000, the R^2 is −0.04% for two slopes, −0.07% for the linear form and the spline, and 0.09% for the network (Panel B of Exhibit 12). Here, statistical and economic significance part in three instructive ways. A statistically clear result can be small: all flexible forms beat the linear forecast in the test of Clark and West (2007) (the network with a t-statistic of 2.7), yet the gain in R^2 is at most 0.12 points, the test of Diebold and Mariano (1995) separates none of them from the linear forecast, superior predictive ability finds no model that beats the linear one (p-value of 0.64), and the model confidence set keeps all (Panel A of Exhibit A22 of the Online Appendix). A large economic result can lack statistical support: the certainty equivalent rises from 2.9% per year for the linear position to 5.2% for the kink and 4.9% for the network, with p-values of 0.49 and 0.15. And the two can agree: the spline and the kernel regression gain 2.8 and 2.1 points, with p-values of 0.02 and 0.04.

The flexible forms thus agree on the asymmetry, though not on the location of the change. The asymmetry is a feature of the data, and once a flexible form has established it, two slopes summarize it: as positions, the spline, the kernel regression, and two slopes span each other, and the network has no alpha once the asymmetric position is added (-0.14% per month, t-statistic of -1.0). We therefore use the two slopes in the remainder of the paper.

5 Two Kinds of Information along the Curve

Having established the shape, we ask where on the curve it lies and whether it merely repeats the current curve. The answer determines how the signal should be traded, in outright positions or in spreads, and what kind of premium it may reflect.

5.1 A One-Sided Level and a Symmetric Spread

Equation (7) splits the return of the nearby contract into the return of the second-nearby contract and the spreading return, long the first- and short the second-nearby contract. Szymanowska et al. (2014) associate nearby returns mainly with the spot premium and spreading returns with term premiums. We use "level" for the return of the second-nearby contract and "spread" for the spreading return.

The level carries the asymmetry (Exhibit 8 and Exhibit 9). Positive basis momentum predicts the second-nearby return as strongly as the nearby return (slopes of 0.061 and 0.059, t-statistics of 6.6 and 6.4), and negative basis momentum predicts neither (-0.005). The difference is larger than for the nearby return (t-statistic of 4.8), and it extends to the third-nearby contract (5.3). In units of the nearby volatility, in which the components add up, the slopes are 0.057 and -0.005 for the level and 0.002 and 0.007 for the spread: the nearby response is almost entirely a response of the level.

The spread does not share the asymmetry. Over its own volatility, its slopes are 0.020

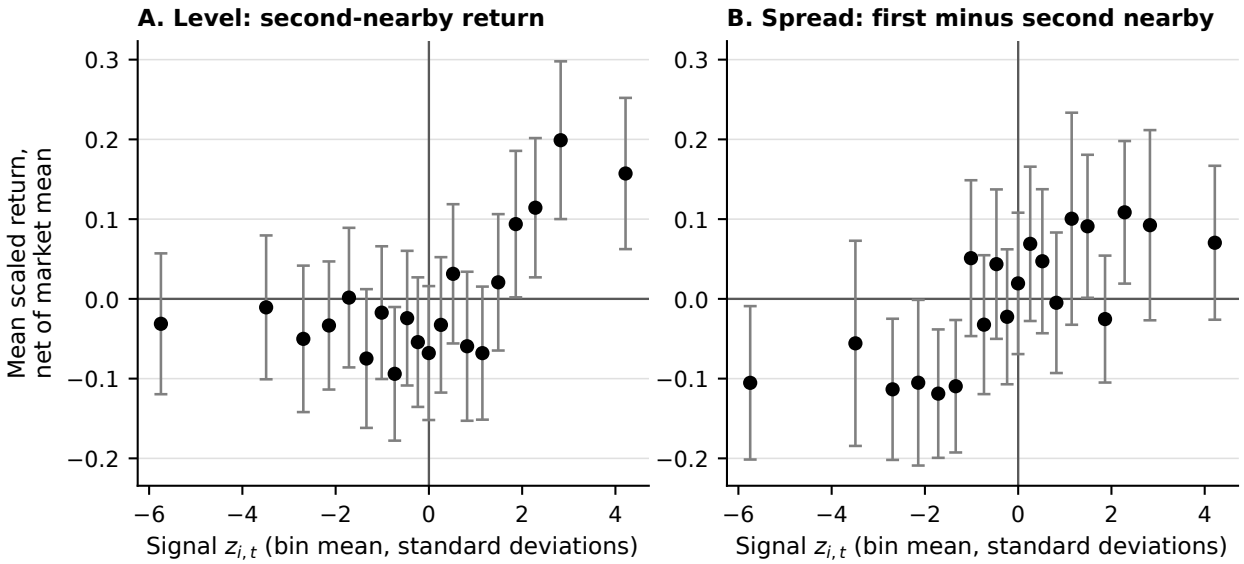


Exhibit 8: **Next-Month Level and Spread by Basis Momentum.** Mean of each return over its own trailing volatility, net of the market's mean, in 20 quantile bins of $z_{i,t}$, 43 markets, July 1961–June 2024, with 95% intervals clustered by month.

Exhibit 9: **Asymmetric Regressions of Nearby, Second-Nearby, and Spreading Returns**

	1960–2024			2000–2024			Kink	
Dependent return	β^+	β^-	$t(\Delta)$	β^+	β^-	$t(\Delta)$	$\hat{\gamma}$	p
Panel A: Asymmetric regressions by return								
Nearby, $r^{(1)}$	0.059***	0.002	4.16	0.040***	−0.011	2.74	−0.10	<0.005
	(6.39)	(0.23)		(3.40)	(−0.95)		[−2.90, 0.95]	
Second nearby, $r^{(2)}$	0.061***	−0.005	4.83	0.049***	−0.016	3.48	−0.10	<0.005
	(6.63)	(−0.68)		(4.12)	(−1.41)		[−1.46, 0.90]	
Spreading, $r^{(1)} - r^{(2)}$	0.020*	0.034***	−0.87	−0.004	0.030**	−1.55	2.60	0.251
	(1.91)	(3.97)		(−0.29)	(2.44)		[−2.95, 3.06]	
Third nearby, $r^{(3)}$	0.063***	−0.010	5.26			3.70		
	(6.73)	(−1.35)						

Panel B: The same comparison across constructions of the signal, 1960–2024

Numerator, denominator	$t(\Delta)$ level	$t(\Delta)$ spread	Contrast	t	Contrast, 2010–24	t
Compounded, 12-month	4.83	−0.87	0.080	4.30	0.065	2.11
Compounded, 24-month	6.59	0.06	0.133	4.61	0.111	1.92
Compounded, 36-month	5.51	0.44	0.111	3.33	0.061	0.89
Compounded, expanding	5.08	−2.00	0.202	5.29	0.150	1.88
Log, 12-month	2.84	0.08	0.031	1.89	0.042	1.59
Log, 24-month	4.21	0.96	0.057	2.06	0.051	0.91
Log, 36-month	3.36	1.78	0.024	0.72	0.019	0.27
Log, expanding	3.18	−0.45	0.111	2.81	0.117	1.35

This table reports Equation (5) for each return over its own trailing 12-month volatility, monthly, 43 markets (third nearby: 41). The Kink columns report Equation (6) with a 90% interval and the sup-F p-value. In Panel B, numerators and denominators are those of Exhibit A12 of the Online Appendix; the contrast is $\beta^+ - \beta^-$ of the level minus that of the spread, stacked. All regressions include market fixed effects; t-statistics are clustered by month. ***, **, and * denote significance at the 1%, 5%, and 10% levels.

(t-statistic of 1.9) and 0.034 (4.0), with no significant difference (-0.9). Negative signals therefore have a role, in the spread and not in outright returns. The asymmetry of the level exceeds that of the spread by 0.080 (t-statistic of 4.3), by 0.099 (4.1) after 2000, by 0.065 (2.1) over 2010–2024, and significantly in six of eight constructions; with a log numerator, it is 0.031 (1.9). The contrast holds with month fixed effects (t-statistic of 4.3), with Driscoll–Kraay standard errors (4.1), and without any one sector (3.3 to 4.8; Panel D of Exhibit A4 of the Online Appendix). These tests locate the predictive information and do not establish its origin. A response that runs along the curve is consistent with a change in expected scarcity, which would affect every contract that matures before new supply arrives.

5.2 Information beyond the Current Basis

Basis momentum accumulates the shape of the curve, so its information could simply repeat the current basis. It does not. With the basis at the time of the signal as a control, the difference between the slopes has a t-statistic of 4.4, with the second spread as well 4.2, and with the trailing means of both spreads 4.2.

Boons and Porras Prado (2019) show that the average nearby return of basis momentum is driven almost completely by roll returns, so we split the return accordingly. The log return of the nearby contract equals the change in the unadjusted first-nearby generic plus the gap between the two contracts at the roll, the form in which Fama and French (1987) study the forecast power of the basis (Equation (A5) of the Online Appendix). Because most generics roll on the first trading days of the delivery month, the roll of month $t+1$ is the basis at the end of month t, known when a position is formed (correlation of 0.95 in the 59% of market-months with a roll).

Without a control, the roll rises with the signal on both sides, more strongly on the positive one (slopes of 0.063 and 0.031, t-statistic of the difference of 4.6). With the basis at t as a control, the roll slopes vanish (-0.002 and -0.003, difference of 0.3), whereas the

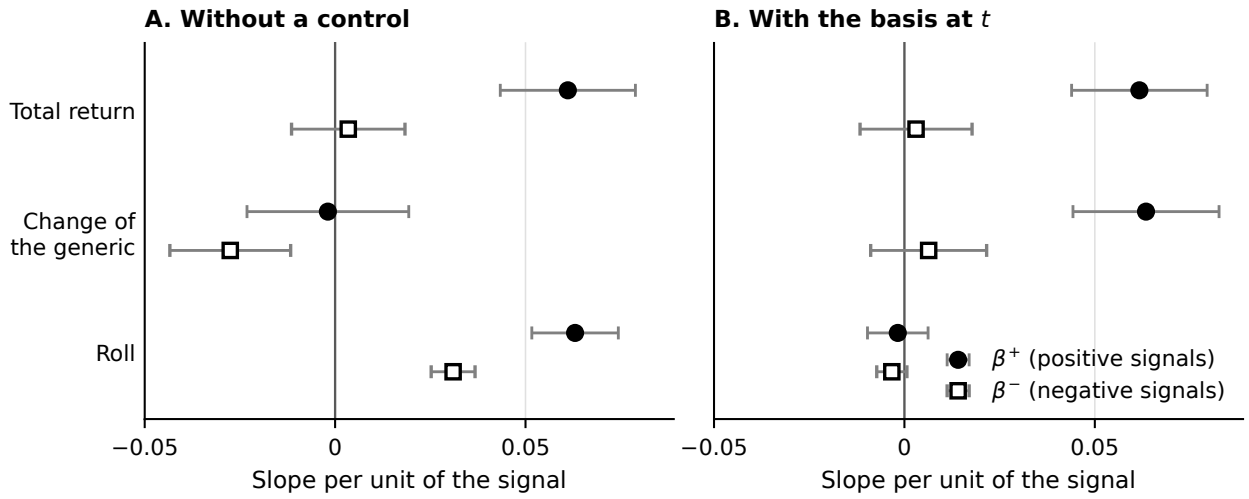


Exhibit 10: **Slopes with and without the Basis at the Time of the Signal.** β^+ and β^- of Equation (5) for each part of the nearby return, 41 markets, July 1961–June 2024, with 95% intervals clustered by month.

total return is unchanged (0.062 and 0.003, difference of 4.4), and the change in the generic shows the same slopes (0.063 and 0.006, difference of 4.1; Exhibit 10). Because the roll of a month is the basis known at its start, the control removes the roll almost by construction; the informative result is that the total-return slopes do not change. The basis itself adds nothing to the total return on either side (−0.015 and 0.015). The slope on the roll thus describes how the signal and today's curve move together, and the information about the return lies outside that curve, on the positive side.

Further cuts agree (Exhibit 11). In months without a roll, where the roll is zero by construction, the asymmetry has a t-statistic of 3.2, which lies mainly before 2000 (1.2 from 2000 on). In the nine markets that roll every month, the total slopes are 0.075 and −0.032 (3.2). In contango, 69% of the observations, the roll shows no asymmetry (−1.1), and the total-return asymmetry (2.5) lies in the generic (2.6). In precious metals, the roll slopes are 0.009 and 0.006, against total slopes of 0.063 and −0.022. The information also lasts. The slope on positive basis momentum is 0.035 per month at twelve months, the difference between the slopes is significant at every horizon up to twelve months, and negative basis momentum is insignificant at every horizon (Exhibit A24 of the Online Appendix). The

Exhibit 11: **The Nearby Return, the Curve at the Time of the Signal, and the Roll**

	Total return			Change of the generic			Roll		
	β^+	β^-	$t(\Delta)$	β^+	β^-	$t(\Delta)$	β^+	β^-	$t(\Delta)$
Panel A: The signal with and without the basis at t as a control									
1960–2024, no control	0.061***	0.003	4.28	−0.002	−0.028***	1.70	0.063***	0.031***	4.56
	(6.75)	(0.45)		(−0.18)	(−3.40)		(10.87)	(10.55)	
1960–2024, basis control	0.062***	0.003	4.36	0.063***	0.006	4.07	−0.002	−0.003	0.35
	(6.80)	(0.41)		(6.48)	(0.82)		(−0.43)	(−1.64)	
1960–1999, basis control	0.085***	0.012	3.59	0.082***	0.009	3.41	0.003	0.003	−0.06
2000–2024, basis control	0.046***	−0.008	2.94	0.052***	−0.002	2.84	−0.006*	−0.006***	0.12
2010–2024, basis control	0.039***	0.016	1.04	0.044***	0.022	0.95	−0.004	−0.006*	0.33
Panel B: Months, markets, and curve states, no control									
Months with a roll	0.050***	−0.003	3.14	−0.051***	−0.062***	0.54	0.101***	0.059***	3.68
Months without a roll	0.080***	0.010	3.18	0.080***	0.010	3.18	0.000	0.000	−0.02
1960–1999	0.125***	0.011	3.58	0.125***	0.011	3.58	0.000	0.000	1.09
2000–2024	0.034*	−0.002	1.21	0.034*	−0.002	1.21	0.000	0.000	−0.32
2010–2024	0.037	0.012	0.63	0.037	0.012	0.63	0.000	0.000	0.50
9 markets that roll every month	0.075***	−0.032	3.24	0.019	−0.057***	2.27	0.056***	0.025***	4.19
Backwardation at t	0.062***	−0.014	2.78	0.018	−0.024	1.37	0.044***	0.010*	2.46
Contango at t	0.048***	0.005	2.45	0.032**	−0.016*	2.64	0.016***	0.021***	−1.07
Month fixed effects	0.047***	0.011*	2.91	−0.014	−0.020***	0.41	0.061***	0.032***	4.20
Panel C: Commodity groups, no control									
Storable	0.064***	0.005	3.87	−0.009	−0.024***	0.89	0.073***	0.029***	5.31
Precious metals	0.063**	−0.022	2.00	0.054**	−0.028	1.91	0.009***	0.006***	0.82
Livestock and dairy	0.046**	0.013	0.85	−0.007	−0.054*	0.94	0.053***	0.067***	−0.44
Panel D: Other constructions of the signal, basis control									
Compounded, 24-month	0.103***	−0.022*	6.20	0.106***	−0.019	5.85	−0.003	−0.003	0.04
Compounded, expanding	0.107***	−0.007	4.01	0.123***	−0.022	4.74	−0.015	0.014***	−2.49
Log, 12-month	0.043***	0.014**	2.53	0.044***	0.018**	2.15	−0.001	−0.004**	0.70
Return volatility, clipped	0.101***	−0.015	3.94	0.118***	−0.018	3.99	−0.017*	0.003	−1.13

Panel E: Mean log return of the nearby contract, % per month, by signal and curve state

	Raw			Net of the market's mean				
Signal, state at t	Total	Generic	Roll	Total	Generic	Roll	Simple	Obs.
$z_{i,t} \leq -2$	−0.31	0.74	−1.05	−0.28	0.50	−0.78	0.05	3,626
backwardation	0.22	−0.92	1.14	0.16	−1.18	1.34	0.72	566
contango	−0.41	1.04	−1.45	−0.36	0.81	−1.17	−0.07	3,060
$-2 < z_{i,t} \leq 0$	−0.28	0.25	−0.54	−0.33	0.00	−0.34	0.07	6,271
$0 < z_{i,t} \leq 2$	0.10	0.09	0.01	−0.01	−0.17	0.16	0.44	5,845
$z_{i,t} > 2$	1.31	0.06	1.25	1.12	−0.20	1.32	1.76	2,928
backwardation	1.78	−0.94	2.72	1.49	−1.20	2.69	2.29	1,657
contango	0.70	1.38	−0.67	0.64	1.11	−0.47	1.06	1,271

This table splits the monthly log return of the nearby contract in the 41 markets with a signal and recoverable roll dates, July 1961–June 2024 or the stated period. Generic is the change of the unadjusted first-nearby generic, which in a month with a roll compares the old with the new contract; Roll is the remainder, so the two add up to the total. Panels A to D report Equation (5) with each part over the trailing 12-month volatility of the nearby return. Basis control adds the positive and negative parts of the basis at t over that volatility. Panel E reports mean log parts, raw and net of each market's mean, and simple means. Market fixed effects; t-statistics clustered by month. Definitions and further statistics are in Section G of the Online Appendix. ***, **, and * denote significance at the 1%, 5%, and 10% levels.

information of positive signals thus concerns the level of the curve and lies outside the current basis, whereas negative signals inform only the spread. This division follows the two predictions of the theory of storage: where the front weakens, carry ties the contracts

together and the signal informs only their spread; where it tightens, the signal also informs the level, as scarcity would. Section 6 asks what this is worth.

6 The Value Lies on the Positive Side

The tests above establish that the two signs differ and where the difference lies. This section measures what the difference is worth in three steps: which side explains the variation of returns, which side adds value for an investor who already holds the commodity premium, and which positions earn abnormal returns with slopes estimated on past data. It closes with the implications for investors and hedgers.

6.1 The Positive Side Explains Most of the Variation

At the median monthly volatility of a market, 6.9%, one unit of a positive signal raises the expected return by 0.40% per month, and one unit of a negative signal changes it by 0.01%. Exhibit 12 reports what each side explains and earns on its own and what it adds to the other. We separate three questions: which side explains statistical variation, which side contributes economic value, and which position earns abnormal returns. The within R^2 of 0.33% is small, as in all return forecasts. By the implied Sharpe ratio of Campbell and Thompson (2008), an out-of-sample R^2 of 0.37% would lift the median market from 0.18 to 0.28, against 0.24 at the linear form's 0.20%, although this gain lies in 1990–1999 (Panel B of Exhibit 12).

The negative side adds nothing to what the positive side explains. Within markets, the R^2 is 0.328% with the positive side alone and 0.329% with both sides, an increment of 0.000 percentage points (jackknife standard error of 0.005), whereas adding the positive side to the negative side raises the R^2 from 0.058% to 0.329% (t-statistic of 3.0). Out of sample, adding the negative side lowers the R^2 from 0.42% to 0.37% (t-statistic of -2.0). By

Exhibit 12: **Attribution of Predictive and Economic Value to the Two Sides**

Panel A: Each side alone, both sides, and the increment of adding each side	Positive alone	Negative alone	Both sides	Adding positive	Adding negative
Statistical					
R^2, in sample, %	0.328	0.058	0.329	0.271	0.000
				(0.089)	(0.005)
R^2, out of sample, %	0.422	−0.002	0.374	0.376	−0.048
				(0.173)	(0.024)
Economic, linear position at 10% volatility					
Sharpe ratio	0.89	−0.18	0.54	0.73	−0.35
Certainty equivalent, % per year	7.10	−4.73	2.91	7.64	−4.20
Economic, opportunity set, squared maximum Sharpe ratio					
With the long-only portfolio	0.93	0.28	0.95	0.67	0.01
				(0.29)	(0.06)
Gain over the six factors	0.27	0.00	0.27	0.27	0.00
				(0.19)	(0.02)

Panel B: Out-of-sample R^2 by period, %	Linear	Two slopes	Spline	Kernel	Network
1990–2024	0.20	0.37	0.31	0.15	0.32
1990–1999	1.14	1.84	1.63	1.46	1.14
2000–2009	−0.25	0.03	−0.03	0.05	0.11
2010–2024	0.04	−0.08	−0.08	−0.38	0.08
2000–2024	−0.07	−0.04	−0.07	−0.22	0.09

Panel A reports each part of the signal, $\max(z, 0)$ or $\min(z, 0)$, alone and both together, and the increment of adding a part to the other, with delete-one-year jackknife standard errors in parentheses. The in-sample value is the R^2 within markets, 1961–2024; the out-of-sample value is that against the prevailing market mean, 1990–2024; the economic values are those of the parts of the linear position, descriptive because the returns retain each market's mean (from February 1993); the squared annual maximum Sharpe ratio, 1990–2024, includes the long-only portfolio in every set, or measures the gain over the six factors of Sakkas and Tessaromatis (2020). Shapley values are in Exhibit A25 of the Online Appendix. Panel B: expanding-window forecasts against the prevailing market mean, re-estimated every eight years, gross of costs, pooled over market-months.

period, the increment of the positive side has t-statistics of 2.5, 1.6, and 1.5 in 1961–1999, 2000–2009, and 2010–2024, and that of the negative side at most 0.9. Shapley values give the same ranking: the positive side accounts for 91% of the explained variation, between 89.7% and 93.2% without any one market and between 87.7% and 94.9% without any one sector, and with momentum, the basis, and hedging pressure as further players, momentum has the largest share (47%), the positive side follows (37%), and the negative side has 2% (Exhibit A25 of the Online Appendix).

The economic values show the same split. At 10% volatility known in real time, the positive part of the linear position earns a Sharpe ratio of 0.89 on its own, the negative part −0.18, and the linear position, which holds both, 0.54: adding the negative side lowers

the Sharpe ratio by 0.35 and the certainty equivalent by 4.2 percentage points per year. These rows are descriptive. The positive part is net long and the negative part net short, so a positive commodity premium separates them even when the two signs carry the same information. The inference therefore rests on benchmarks that hold the premium. In the opportunity set with the long-only portfolio, adding the positive side raises the squared maximum Sharpe ratio from 0.28 to 0.95 (a gain of 0.67, t-statistic of 2.3), and adding the negative side raises it from 0.93 to 0.95 (0.01, standard error 0.06). Beyond the six factors, whose squared maximum Sharpe ratio is 1.45, the positive side adds 0.27 (t-statistic of 1.4) and the negative side nothing. The negative-signal position shows no statistically detectable average premium in these specifications, while negative signals keep their role in the spread (Section 5).

6.2 Investment Value of an Asymmetric Position

We compare linear basis momentum with the asymmetric position, whose slopes are estimated on past data, and with the positive-only rule, which requires no estimation (Section 2.4). The split at zero is chosen in knowledge of the shape, and we re-estimate on expanding windows with data through December 1989, 1997, 2005, 2013, and 2021 (walk-forward, February 1990–June 2024).

Over 1990–2024, the asymmetric position earned an alpha of 0.30% per month gross of costs on linear basis momentum and the long-only portfolio (t-statistic of 3.0), at a volatility of 29% per year, or about 1.2% per year at 10% volatility; the positive-only rule earned 0.33% (3.3) (Exhibit 13). Because the position is net long, the fair raw benchmark also holds the commodity premium: its Sharpe ratio of 0.86 exceeds the 0.75 of a mix of linear basis momentum and the long-only portfolio, scaled ex post (one-sided p-value of 0.02). With mix weights known in real time, from 1993, the gain is not significant (0.82 against 0.75, p-value of 0.19), and from 2000 neither. On the six factors with linear basis momentum

Exhibit 13: **Walk-Forward Performance of Time-Series Strategies**

	Linear BM	Asymmetric	Neural net	Positive-only	Long-only	TS momentum
Walk-forward, February 1990–June 2024						
Sharpe ratio	0.63***	0.86***	0.78***	0.85***	0.43**	0.67***
Δ Sharpe vs. linear BM		0.23*	0.15	0.22	−0.20	0.04
α on linear BM, long-only		0.30***	0.26	0.33***		
		(3.01)	(1.49)	(3.26)		
α on linear BM, long-only, TS momentum		0.24**	0.11	0.28***		
		(2.35)	(0.63)	(2.76)		
α on the above and cross-sectional factors		0.25**	0.10			
		(2.36)	(0.57)			
Loading on long-only, first α regression		0.91	0.51			
Volatility (% per year)	23	29	26	20	19	14
α on linear BM, long-only, TS momentum, asymmetric			−0.14			
			(−0.97)			
Single split, February 2000–June 2024						
Sharpe ratio	0.40*	0.77***	0.23	0.83***	0.57**	0.44**
Δ Sharpe vs. linear BM		0.37**	−0.17	0.43*	0.17	0.04
α on linear BM, long-only		0.31***	0.32*	0.32***		
		(2.81)	(1.86)	(2.81)		
α on linear BM, long-only, TS momentum		0.29**	0.31*	0.30**		
		(2.56)	(1.75)	(2.56)		
α on the above and cross-sectional factors		0.29**	0.31*			
		(2.50)	(1.68)			
Loading on long-only, first α regression		0.83	−0.52			
Volatility (% per year)	23	29	23	21	20	14
α on linear BM, long-only, TS momentum, asymmetric			−0.04			
			(−0.51)			

This table reports walk-forward strategies gross of costs (Equation (9)), monthly, 43 markets, all on the same markets and months. Positions are defined in Section 6.2, estimated with data through December 1989, 1997, 2005, 2013, and 2021 (single split: 1999). Sharpe ratios are annualized, with one-sided bootstrap stars. α is in % per month, with Newey–West t-statistics (6 lags) in parentheses. Factors and further statistics are in Section G of the Online Appendix. ***, **, and * denote significance at the 1%, 5%, and 10% levels.

added, the alpha is 0.28% (t-statistic of 2.0). Estimating the location in real time as well gives t-statistics of 3.3 with the estimator of the position and 1.9 with market fixed effects; with market fixed effects, the alpha is 0.12% per month through 2005 (t-statistic of 1.0) and 0.54% (2.2) from 2006, because the location estimated in real time lies near −3 until the late 1990s. Charging both legs of every roll at 5 and 10 basis points lowers the Sharpe ratio to 0.75 and 0.65, with the gain over the mix unchanged (Exhibit A26 of the Online Appendix), and the positive-only rule is sturdier when notional per market is capped: at caps of 10 and 5 times a market's share of capital, its alpha is 0.27% and 0.18%, against 0.14% (t-statistic of 1.9) and 0.09% (1.4) for the position. These alphas also depend on the measurement of the signal. With signal and return both under the contract rule of Boons and Porras Prado (2019), the slopes still differ (t-statistic of 3.9), and the alpha of the position has a t-statistic

of 0.7; with the generic signal and the return earned on those contracts, it stays between 3.2 and 3.8 (Panel D of Exhibit A8 of the Online Appendix). Section 7.2 reports signals that leave out the last days before the roll. The main risk is a commodity crash (Panel A of Exhibit A27 of the Online Appendix).

6.3 Sizing Adds Value beyond Direction

Exhibit 14 compares the asymmetric position with binary rules, following the comparison of linear, binary, and nonlinear weights in Moskowitz et al. (2026). These rules make the same sign calls without sizing positions, so they separate direction from size.

The sign of basis momentum (the binary rule of Moskowitz et al., 2012) earns a Sharpe ratio of 0.58 in the walk-forward period, less than linear basis momentum, and the efficient weight of Ferson and Siegel (2001), which shrinks extreme signals, earns less as well. The asymmetric position beats every rule in the walk-forward period, with an alpha of 1.15% per month (t-statistic of 3.0) on the sign rule and of 0.82% (2.8) on the long-or-flat rule, and the positive-only rule, which makes the same sign calls as the long-or-flat rule, earns 0.43% on it (3.9). The convexity coefficients θ of Henriksson and Merton (1981) and Treynor and Mazuy (1966), which measure timing of the market's direction (Merton, 1981), are insignificant at the 5% level for all positions in these regressions, which do not hold linear basis momentum. Inside the regression of the alpha, Equation (10), the tests do detect convexity: with the squared commodity return added to linear basis momentum and the long-only portfolio, the alpha falls from 0.30% to 0.11% (t-statistic of 1.1), with a coefficient of 0.60 (t-statistic of 2.8), and with the Henriksson–Merton term, it falls to 0.01% (0.04). The coefficient lies in the crash years: without 2008, 2009, and 2020, it has a t-statistic of 1.0, while the alpha stays at 0.14% (1.1). The convex terms are not traded returns, so these intercepts are not alphas: by the equivalence of Merton (1981), a convex payoff obtained from a known signal is a form of timing value. Against time-series momentum, a traded strategy with a convex

Exhibit 14: **Binary Benchmarks and Market-Timing Regressions**

Panel A: Sharpe ratios

	Walk-forward, 1990–2024			Single split, 2000–2024		
	Sharpe	Asym. − row	Turnover	Sharpe	Asym. − row	Turnover
Linear, $f(z) = z$	0.63***	0.23*	1.34	0.40*	0.37**	1.26
Binary, $f(z) = \text{sign}(z)$	0.58***	0.28**	0.56	0.29	0.48***	0.52
Long-or-flat, $f(z) = \mathbb{1}\{z > 0\}$	0.64***	0.22*	0.27	0.62***	0.15	0.26
Ferson–Siegel, $f(z) = z/(1+z^2)$	0.53***	0.34**	0.27	0.31*	0.46**	0.25
Asymmetric	0.86***		1.35	0.77***		1.29
Neural network	0.78***	0.09	1.48	0.23	0.54**	1.21

Panel B: Alpha of the asymmetric position on the binary benchmarks, % per month

	Walk-forward, 1990–2024		Single split, 2000–2024	
Binary	1.15***	(3.01)	1.38***	(3.03)
Binary, long-only, TS momentum	0.51**	(2.35)	0.60***	(2.70)
Long-or-flat	0.82***	(2.82)	0.63*	(1.94)
Long-or-flat, TS momentum	0.72***	(2.65)	0.63**	(2.03)
Positive-only rule on long-or-flat	0.43***	(3.86)		

Panel C: Market timing, walk-forward, 1990–2024

	Henriksson–Merton θ		Treynor–Mazuy θ		Market loading	
	Portfolio	Per market	Portfolio	Per market	Portfolio	Per market
Linear, $f(z) = z$	0.045	0.219	0.214	0.106	−0.03	−0.23
	(0.17)	(1.29)	(0.26)	(0.38)		
Binary, $f(z) = \text{sign}(z)$	−0.017	0.012	−0.039	−0.078	0.03	−0.02
	(−0.17)	(0.16)	(−0.12)	(−0.64)		
Long-or-flat, $f(z) = \mathbb{1}\{z > 0\}$	−0.008	0.006	−0.020	−0.039	0.51	0.49
	(−0.17)	(0.16)	(−0.12)	(−0.64)		
Ferson–Siegel, $f(z) = z/(1+z^2)$	0.011	−0.002	0.041	−0.033	0.00	0.00
	(0.33)	(−0.09)	(0.39)	(−0.69)		
Asymmetric	0.183	0.321*	0.810	0.197	0.80	0.64
	(0.66)	(1.78)	(0.90)	(0.63)		
Neural network	0.228	0.241	0.898	0.094	0.38	0.23
	(0.82)	(1.40)	(0.86)	(0.33)		

Panel D: Alpha of the asymmetric position on linear BM and long-only, with convexity in the long-only return, walk-forward

	Base	Henriksson–Merton		Treynor–Mazuy	
	α	α	θ	α	θ
1990–2024	0.30***	0.01	0.14*	0.11	0.60***
	(3.01)	(0.04)	(1.73)	(1.09)	(2.75)
1990–2009	0.42***	−0.13	0.27***	0.15	0.94***
	(3.34)	(−0.71)	(3.59)	(1.21)	(4.43)
2010–2024	0.17	0.18	0.00	0.09	0.25
	(1.19)	(0.61)	(−0.03)	(0.49)	(0.63)
Without 2008, 2009, 2020	0.24**	0.08	0.08	0.14	0.42
	(2.52)	(0.42)	(0.90)	(1.07)	(1.05)

This table reports binary benchmarks and timing tests for the markets and months of Exhibit 13, February 1990 or 2000–June 2024. Panel A: Sharpe ratios with one-sided bootstrap stars; Asym. − row is the difference to the row. Panel B: α of the asymmetric position on the benchmarks, in % per month. Panel C: timing regressions of Henriksson and Merton (1981) and Treynor and Mazuy (1966), for the portfolio and per market. Newey–West t-statistics, clustered by month per market. Rules and equations are in Section G of the Online Appendix. ***, **, and * denote significance at the 1%, 5%, and 10% levels.

payoff (Moskowitz et al., 2026), the alpha is 0.24% (t-statistic of 2.3; Section E of the Online Appendix). A position that responds to how far a signal lies above zero earns more than one that responds only to its sign, which is the portfolio counterpart of the slope in Exhibit 3.

6.4 Implications for Investors and Hedgers

The results bear on four uses of the signal. For an investor in outright futures, the information lies on one side. At the median volatility of a market, a signal of two raises the expected return by about 0.8% per month, and markets with signals above two earn 1.09% per month more than their own average (t-statistic of 4.4), whereas a signal of minus two changes it by almost nothing. A short position taken because basis momentum is negative therefore carries the costs and the volatility of a trade without a detectable premium: for an investor who already holds the commodity premium, adding the negative side raises the squared maximum Sharpe ratio by 0.01, against 0.67 for the positive side. Leaving it out, as the positive-only rule does, halves turnover relative to linear basis momentum and keeps an alpha of 0.33% per month (t-statistic of 3.3; Exhibit A26 of the Online Appendix).

For a trader of calendar spreads, the conclusion differs. The spread between the first two contracts responds to both signs, with slopes of 0.020 and 0.034 that do not differ significantly, so the data do not reject a linear use of the signal in spread positions. The same signal thus calls for a one-sided position in outright returns and a two-sided position in spreads, and a single rank applied to both cannot reflect both.

For hedgers, who are mostly short futures, a positive signal predicts a higher return on each of the first three contracts and hence a higher expected cost of a short hedge over the following months, and commercial hedgers raise their net short positions in the following months (Section 7.3). A negative signal does not predict a lower cost. For asset pricing, the factor built from the rank of basis momentum combines a long leg that carries the premium with a short leg whose return does not reflect negative signals, so that tests of what the factor prices, and of its source, should be run by sign.

The economic scale is modest but not negligible. The within R^2 of 0.33% is small, as in all return forecasts, yet by the implied Sharpe ratio (Section 6.1), an out-of-sample R^2 of

0.37% raises the attainable Sharpe ratio of the median market from 0.18 to 0.28. After roll costs of 5 and 10 basis points on both legs, the asymmetric position keeps Sharpe ratios of 0.75 and 0.65. These gains lie mainly before 2010, and since 2010 the alpha is 0.17% per month (t-statistic of 1.2), so they measure the historical value of the information rather than a forecast of future profits. Whether that value reflects a premium or information that the market absorbs slowly is the subject of Section 7.

7 Bounds on the Source of the Asymmetry

The evidence so far is predictive. This section confronts it with candidate accounts: conventional explanations that could produce a one-sided pattern mechanically, the behavior of the curve and of the return distribution after positive signals, risk, price pressure, and hedging demand, and finally the weakening of the asymmetry since 2010.

7.1 Conventional Accounts Fail

Three conventional accounts fail on their own terms. The commodity premium would make the long side of any time-series strategy dominate (Huang, Li, Wang, and Zhou, 2020), but the asymmetry survives market fixed effects, and no other commodity signal shares it under all scalings. A symmetric S-shape, which follows from portfolio theory if expected returns are linear in the signal (Moskowitz et al., 2026), is rejected, since the two branches differ. Scarcity as a current state, in which a tight curve sharpens the positive side, is also not necessary: the slope on positive basis momentum is 0.062 (t-statistic of 4.4) in backwardation and 0.049 (3.9) in contango, it does not rise with the basis as a continuous variable (-0.4), and it is 0.063 in precious metals. Exhibit 15 reports tests that separate the remaining accounts. The tests and their readings were fixed before the first run. We added afterward the sum of the two volatility slopes, known volatility as a control, the slopes within market groups, and

the rows with flexible controls in Panel A; the move into backwardation is in the plan and outside the set of tests named for a joint adjustment.

7.2 Tightening Curves and Upside Returns after Positive Signals

Positive signals are followed by a higher probability of backwardation. Panel A of Exhibit 15 shows that, among observations in contango and with a linear control for the current basis, one unit of a positive signal is associated with a probability of backwardation higher by 1.4 percentage points a month later and by 2.1 points six months later, with differences from the negative side that have t-statistics of 3.1, 2.5, and 2.5. Crossing zero depends strongly on the distance from it, so the panel also reports flexible controls, added after the first run. With 20 bins of the distance, the positive slopes are 0.5, 1.1, and 1.4 points at one, three, and six months (t-statistics of 1.7, 2.4, and 2.5), and the differences have t-statistics of 1.8, 1.9, and 2.0. With the curve's trailing history as well, the positive slopes are no longer significant, the negative slopes turn significantly negative at these horizons, and the differences stay near two. The mean change in the basis also responds on both sides, more strongly on the positive one at one month (2.3), with weaker differences at three and six months (1.8 and 1.7), and the difference at twelve months, the horizon we named in advance, is zero (0.2). We therefore read this as a supporting pattern: a market need not be backwardated when the signal is positive, since a higher probability of tightening follows the signal.

The return distribution widens on the upside. Panel B shows that positive signals leave the lowest decile unchanged (slope of 0.014 at the 10th percentile, t-statistic of 0.9), raise the median (0.046, 4.1), and raise the upper tail most (0.103 at the 90th percentile, 5.6). Negative signals move no quantile significantly. The response at the median shows that the premium does not require the rare spikes that models of competitive storage emphasize (Deaton and Laroque, 1992, 1996). A tightening curve and a wider upper tail after positive signals are the two patterns that a convex convenience yield predicts.

The last days before the roll bear on the alpha of the estimated position and not on the asymmetry. Generics hold the nearby contract until the delivery month. Panel E splits the generic signal into the signal on contracts left 3, 5, and 10 trading days earlier and the remainder from the last days. The remainder has a significant slope on the positive side in each case (0.041, 0.045, and 0.033; t-statistics of 3.1, 4.1, and 3.2) and none on the negative side, beyond the earlier signal. The earlier signal keeps the larger slopes and its own asymmetry, and the asymmetry of the remainder is significant at five days (t-statistics of 1.5, 2.4, and 1.1). On the 41 markets with roll dates, a position built on the earlier signal, with the return still earned on the generic, has an alpha of 0.21%, 0.06%, and 0.18% per month (t-statistics of 1.7, 0.6, and 1.4 on its own linear benchmark), against 0.33% (3.3) for the generic signal. A true leave-out signal, which does not accumulate the spread in the last 1 to 15 days, gives the same picture for the position, with t-statistics from 0.5 to 2.5, and a difference from the generic position that is significant only at five days. The positive-only rule, which estimates nothing, keeps a significant alpha on such signals at 10 of 11 horizons (t-statistics from 2.3 to 3.2, and 1.7 at five days), and the slopes differ at all of them (2.7 to 5.0). The last days thus carry part of the linear premium and of the alpha of the estimated position. The asymmetry keeps its sign and stays significant at the 5% level when both legs leave the expiring contract 1 to 15 days early, but at five and six days it falls below our hurdle (t-statistics of 2.2 and 2.4), and before 2000 it is between 1.0 and 3.5 under early rolls and 1.3 under the contract rule of Boons and Porras Prado (2019), against 3.4 on the generics of the same markets (Panels C and D of Exhibit A13 of the Online Appendix).

7.3 Evidence against Volatility and Temporary Pressure

Volatility does not separate the two sides. Panel C shows that realized volatility rises with the size of the signal on both sides of zero, by similar amounts per unit (0.010 and −0.011 on z^+ and z^-; t-statistic of −0.1 for equal size, a test added after the first run), and about

half of this rise is mean reversion of volatility (Section F of the Online Appendix). Negative signals thus carry as much added volatility as positive signals and no added return, so that volatility alone does not explain a one-sided return. Risk more broadly remains possible, because positive signals do raise volatility. With the commodity return of the next month and its interactions, the difference between the slopes falls from 0.057 to 0.032 and remains significant (t-statistic of 2.8), and the slope on positive basis momentum stays at 0.042 (5.4). The fall mainly reflects the commodity return that positive signals forecast. The interaction with z^+, the test of a higher commodity beta, is insignificant (1.6). With month fixed effects, the slope is 0.046 (Exhibit 3).

No reversal of the gains is detected, which counts against temporary price pressure. Panel D shows that the 12-month gain after positive signals (0.420 per unit, t-statistic of 4.5) is not given back in months 13–24 (0.010, 0.1) or 25–36 (0.065, 0.9), although the interval for months 13–24 allows a partial reversal. On the negative side, the slope in months 25–36 is −0.179 (−2.0). This leaves a premium or slow-moving information. Hedging demand is a candidate for the premium. Positive signals are followed by a rising net short position of commercial hedgers at one and three months (differences of 2.4 and 2.0), which does not last to twelve months (0.3), and the positive-side slope is 0.060 when hedgers are net short, against 0.023 otherwise (Exhibit A29 of the Online Appendix), a difference of 0.037 with an interval from −0.011 to 0.085; within markets, the slopes are 0.050 and 0.039. Producers also hedge more into rising prices, however, so these patterns do not establish the direction of the effect. With a high VIX, the positive-side slope rises by 0.044 (t-statistic of 2.1), in line with limited capacity of speculators (Cheng, Kirilenko, and Xiong, 2015; Acharya, Lochstoer, and Ramadorai, 2013), although none of about 13 such interactions survives an adjustment for multiplicity (Section F of the Online Appendix).

Across the 14 tests named in advance, a joint Holm adjustment retains the rise in volatility with the signal and the difference at the 90th percentile. The fall in the positive slope since 2010 has an adjusted p-value of 0.07, and the other tests have 0.15 or more, including the

change in the basis at twelve months (1.0); the move into backwardation was not among them. The evidence therefore bounds two accounts: it counts against volatility alone and temporary pressure as explanations. It does not separate a scarcity premium, a hedging premium, and slow information, and the source remains open. The results are consistent with scarcity not yet fully reflected in the curve, but equal slopes in contango and backwardation and in precious metals do not favor it, the asymmetry is not confined to the part of the signal specific to a market, and we measure inventories only through the curve. Boons and Porras Prado (2019) argue against storage because basis momentum also predicts currencies and stock and bond indexes, and Gu et al. (2025) confirm this for financial futures. We do not test the two signs there; a one-sided response in those markets would count against the scarcity reading.

7.4 A Weaker Asymmetry since 2010

On the generics, the asymmetry is strong through 2009 and is not significant afterward in the pooled sample (Exhibit 3, last column, and Exhibit A14 of the Online Appendix). It holds on both sides of the dates at which the literature locates a weakening of commodity signals, December 2000 for momentum (Hollstein, Prokopczuk, and Tharann, 2021) and 2004 (Baba-Yara and Bondatti, 2023), with t-statistics of 2.6 and 2.3 afterward. An estimated break selects March 2009, a date of Bianchi, Fan, Miffre, and Zhang (2023), with a p-value of 0.16 after the search over dates (Panel F of Exhibit A13 of the Online Appendix). The timing points away from publication, without excluding it: the difference is already -0.01 from 2010 to April 2015, before the first working-paper version of Boons and Porras Prado (2019) (Panel C of Exhibit A28 of the Online Appendix). Over 2010–2024, the difference between the slopes is 0.018, with a 95% interval from -0.025 to 0.061 that contains the full-sample value of 0.057.

The change lies on the positive side. From 2010, the positive slope falls by 0.051 (t-

statistic of −2.8), whereas the negative slope does not change (0.7), and large signals have not become rarer: a signal above two occurs in 19% of the observations since 2010, against 15% before 2000. Panel F shows that, in point estimates, the fall lies in the markets of the S&P GSCI, where the positive slope drops from 0.088 to 0.012 and the difference from a t-statistic of 4.2 to −0.4. In the other markets, the slopes are nearly unchanged, at 0.060 and −0.001 through 2009 and 0.052 (t-statistic of 2.3) and −0.006 since 2010, with a difference of 0.058 (t-statistic of 1.8); over 2010–2019, the positive slope in these markets is 0.041 (1.7). The entry of financial investors into index markets (Cheng et al., 2015; Baker, 2021) is one possible interpretation, not one we test: the difference between the two groups in the shift is not significant (t-statistic of −1.5, and −0.2 at 2004), and the asymmetry stays significant after 2004. The weakening thus changes the reading of the result from a timeless trading rule to a historically important feature of the signal. The alpha of the asymmetric position is 0.42% per month through 2009 (t-statistic of 3.3), and since 2010, it is distinguishable neither from zero (0.17%, t-statistic of 1.2) nor from its earlier level (shift of −0.25%, t-statistic of −1.3). With 24- and 36-month volatilities, the difference over 2010–2024 has t-statistics of 3.2 and 3.0, which come from 2020–2024 (Exhibit A12 of the Online Appendix).

Exhibit 15: **Tests of Candidate Accounts of the Asymmetry**

	z^+	t	z^-	t	$t(\Delta)$
Panel A: The curve after the signal, with the basis at t as a control					
Into backwardation from contango, 1 month	0.014	(4.27)	0.000	(0.18)	3.12
Into backwardation from contango, 3 months	0.019	(3.94)	0.003	(1.16)	2.50
Into backwardation from contango, 6 months	0.021	(3.52)	0.000	(−0.02)	2.47
Into backwardation from contango, 12 months	0.001	(0.18)	−0.014	(−2.60)	1.69
20 bins of the distance from zero, 1 month	0.005	(1.67)	−0.002	(−1.30)	1.83
20 bins of the distance from zero, 3 months	0.011	(2.45)	0.000	(0.05)	1.86
20 bins of the distance from zero, 6 months	0.014	(2.47)	−0.002	(−0.65)	2.04
With the curve's history as well, 1 month	0.002	(0.77)	−0.006	(−3.16)	2.03
With the curve's history as well, 3 months	0.006	(1.35)	−0.006	(−2.42)	2.15
With the curve's history as well, 6 months	0.008	(1.38)	−0.010	(−2.81)	2.31
Change in the basis, 1 month	0.024	(3.76)	0.008	(2.50)	2.30
Change in the basis, 3 months	0.037	(4.37)	0.017	(3.78)	1.80
Change in the basis, 6 months	0.038	(3.85)	0.015	(2.24)	1.71
Change in the basis, 12 months	−0.010	(−1.37)	−0.012	(−1.65)	0.24
Panel B: Quantiles of the return					
Quantile 0.10	0.014	(0.88)	0.011	(0.94)	0.15
Quantile 0.25	0.044	(4.44)	0.014	(1.70)	1.87
Quantile 0.50	0.046	(4.10)	0.005	(0.70)	2.46
Quantile 0.75	0.064	(6.35)	−0.003	(−0.37)	4.33
Quantile 0.90	0.103	(5.60)	−0.024	(−1.40)	4.29
Panel C: Risk					
Log realized volatility, net of known volatility	0.010	(3.53)	−0.011	(−4.83)	4.93
Return, with the commodity return and interactions	0.042	(5.43)	0.010	(1.54)	2.85
Interaction with the commodity return	0.025	(1.59)	−0.016	(−0.93)	
Panel D: Hedgers and reversal					
Change in hedging pressure, 1 month	0.004	(4.52)	0.001	(0.67)	2.42
Change in hedging pressure, 3 months	0.005	(3.04)	0.000	(0.26)	1.96
Change in hedging pressure, 12 months	0.001	(0.41)	0.000	(−0.01)	0.32
Sum of returns, months 1–12	0.420	(4.49)	−0.034	(−0.38)	3.17
Sum of returns, months 13–24	0.010	(0.11)	−0.039	(−0.52)	0.37
Sum of returns, months 25–36	0.065	(0.91)	−0.179	(−2.03)	1.87
Panel E: Return, signal split at the last days before the roll					
Signal on contracts left 3 days earlier	0.054	(5.74)	0.012	(1.48)	3.08
Remainder, last 3 days	0.041	(3.09)	0.012	(0.94)	1.50
Signal on contracts left 5 days earlier	0.048	(4.86)	0.016	(2.04)	2.23
Remainder, last 5 days	0.045	(4.08)	0.006	(0.53)	2.43
Signal on contracts left 10 days earlier	0.058	(5.64)	0.015	(2.00)	3.04
Remainder, last 10 days	0.033	(3.17)	0.014	(1.19)	1.09
Panel F: Return by market group and period					
S&P GSCI, 1961–2009	0.088	(5.89)	−0.002	(−0.16)	4.24
S&P GSCI, 2010–2024	0.012	(0.68)	0.024	(1.11)	−0.37
Other markets, 1961–2009	0.060	(2.70)	−0.001	(−0.08)	1.91
Other markets, 2010–2024	0.052	(2.26)	−0.006	(−0.37)	1.81

This table reports regressions on $z^+_{i,t}$ and $z^-_{i,t}$ with market fixed effects, monthly, up to 43 markets, July 1961–June 2024; hedging pressure from 1986. $t(\Delta)$ is the t-statistic of the difference. Panel A: an indicator of backwardation at $t+h$ for observations in contango at t, and the change in the basis over known return volatility; the rows with 20 bins of the basis and of the scaled basis, and with its trailing 12-month mean and share of backwardation, were added after the first run. Panel B: quantile regressions of the return net of the market's mean, with a month bootstrap (199 draws). Panel C: daily realized volatility of $t+1$, and the return with the equal-weight commodity return of $t+1$ and its interactions. Panel E: 41 markets; the generic signal split into the signal on earlier contracts and the remainder, both over the volatility of the generic signal, with the positive and negative parts of each. Panel F: within the 24 markets of the S&P GSCI and the other markets. t-statistics are clustered by month at one month and Driscoll–Kraay with bandwidth h at horizon h.

8 Conclusion

This paper shows that basis momentum carries two kinds of information. Over 1960–2024, both signs predict the spread between the first two contracts with similar slopes, whereas only positive signals predict the level of the curve: positive signals predict the returns of the first three contracts (slopes of 0.059, 0.061, and 0.063), and negative signals, on average over the sample, predict none of them (0.002 on the nearby contract, t-statistic of 0.2). In outright returns, predictability is concentrated in positive signals, and the difference between the slopes clears the hurdle of three and survives controls for momentum, the basis, and hedging pressure.

Flexible estimators recover this shape rather than manufacture it. A neural network whose specification contains no split, kink, or number of changes in slope recovers the one-sided shape, rejects a linear truth, and reports no asymmetry when the truth is linear or a symmetric S-shape. A spline and a kernel regression agree on the asymmetry, and two slopes capture what the flexible forms find, so that the network validates a simple specification. The asymmetry lies in the level of the curve rather than in calendar spreads, and it is not a repackaging of the current basis.

Economically, the negative side adds nothing to the variation explained within markets once the positive side is known and 0.01 to the squared maximum Sharpe ratio, against 0.67 for the positive side, and short positions taken solely on negative signals earn no premium. With the split at zero, a position with separate slopes earned a gross alpha of 0.30% per month over 1990–2024 on the generic signal, and 0.24% against time-series momentum as well. A symmetric position thus combines a positive exposure that carries the value with a short exposure that earns no corresponding premium. The implications follow from where the information lies. Investors in outright futures gain little from shorting on negative signals and can leave them out at half the turnover; traders of calendar spreads can use both signs;

hedgers face a higher expected cost of a short hedge after positive signals and no lower cost after negative ones; and factor models that rank basis momentum combine a priced long leg with a short leg that carries no corresponding information.

The evidence is inconsistent with volatility alone or temporary price pressure and fits scarcity that the curve does not yet reflect. Since 2010, the contrast between level and spread has persisted (t-statistic of 2.1), whereas the difference between the outright slopes has narrowed (0.8).

These results change the interpretation of basis momentum from a symmetric long-short signal to a predominantly one-sided predictor of commodity returns. Tests of its source should therefore be run by sign, in commodity markets with direct measures of inventories and in financial futures, where the split by sign gives a direct test of the storage channel.

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